Shrinkages of Concrete

Interpretation-Simulation-Regulation

For GRC Advanced Materials for Sustainable Infrastructure Development, *3-8, August, 2014, HK, China*

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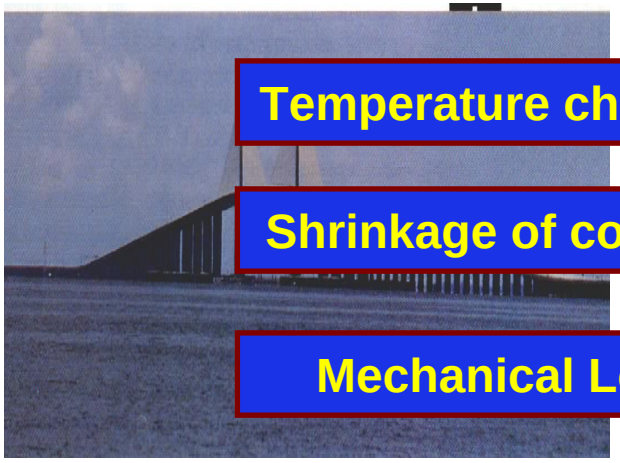
1. INTRODUCTION

- **Damage Processes of General Concrete Structures**

Environmental Action I
(1. Heating-Cooling、Wetting-Drying
2. Mechanical Load)

Environmental Action II
Water and chemical agents (O_2, CO_2, Cl^-, SO^{4-}) come in through the cracks

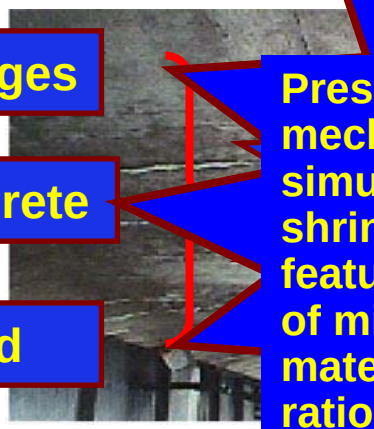
New cast concrete structures → Formation of initial macro or micro-cracks → Corrosion of steel; Delamination of concrete cover; Stop service



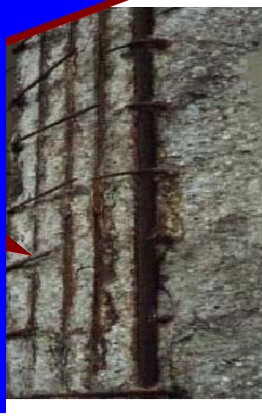
Temperature changes

Shrinkage of concrete

Mechanical Load



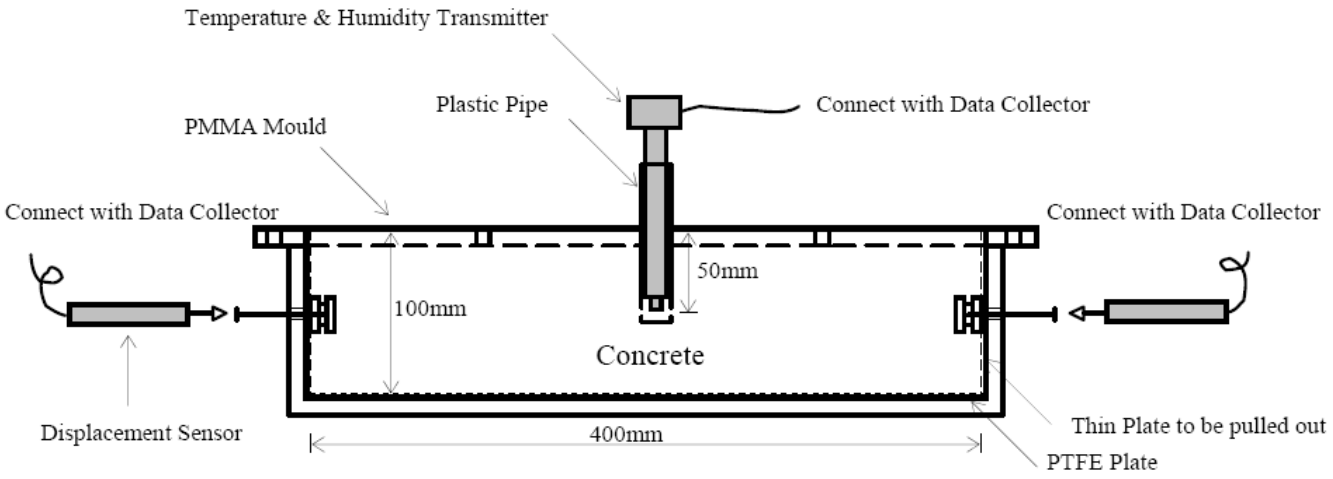
Present talk will be focus on the mechanism interpretation, simulation and regulation of shrinkage modern concrete, that featured by relative large amount of mineral supplementary materials and low water-binder ratio.

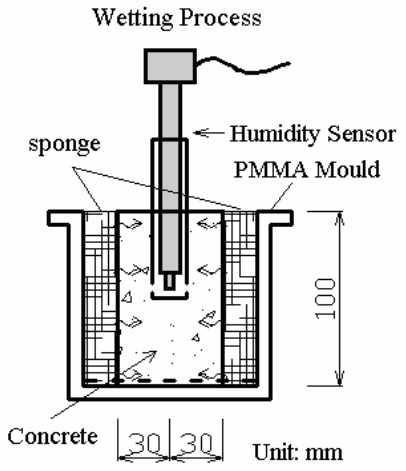


2. MECHANISMS

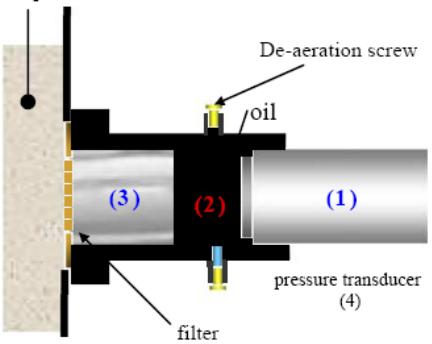
Experimental Observations

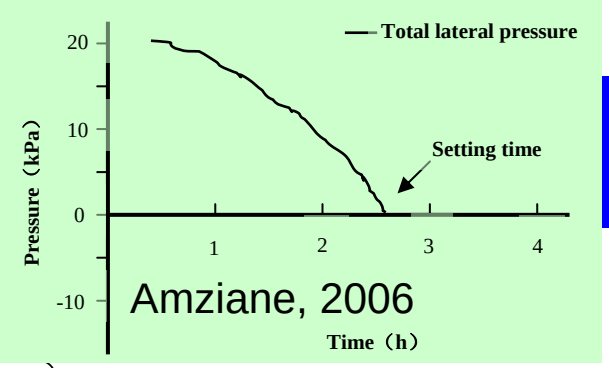
- The mold used was made of plexiglass with inner dimensions of 60×100×400 mm. **The four inner sides of the mold were covered with four pieces of removable plastic sheets** with 2 mm thickness and the bottom of the mold was covered with a thin vinyl sheet with 1 mm thickness to reduce the frictional resistance between the mold and the concrete. After initial set of the concrete, the inner four removable plastic sheets were lifted to create the "**free restraint**" conditions of the shrinkage test.
- The humidity and temperature at the center of the test specimen were measured. The deformation was measured by two linear variable differential transducers (LVDTs).

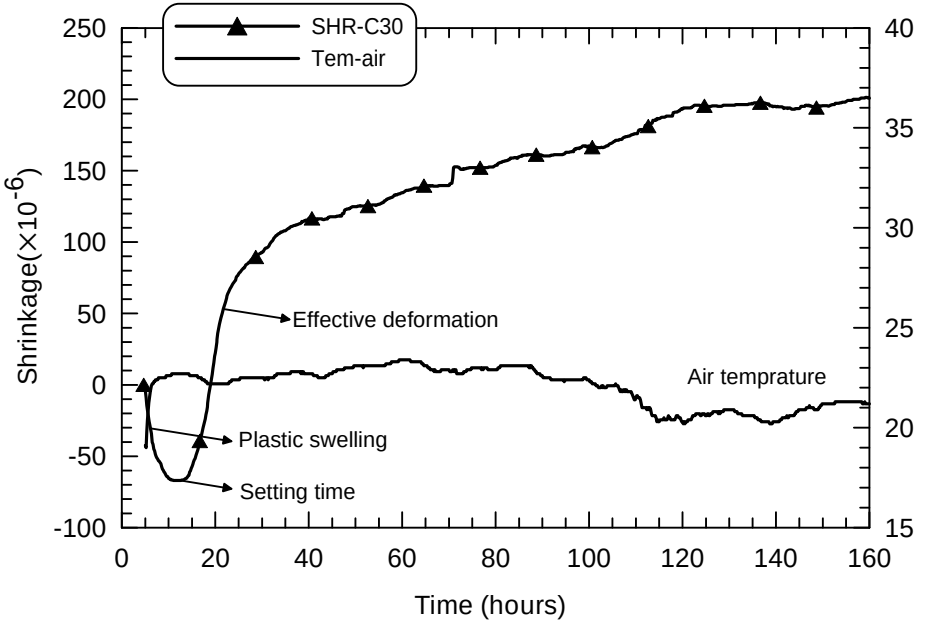


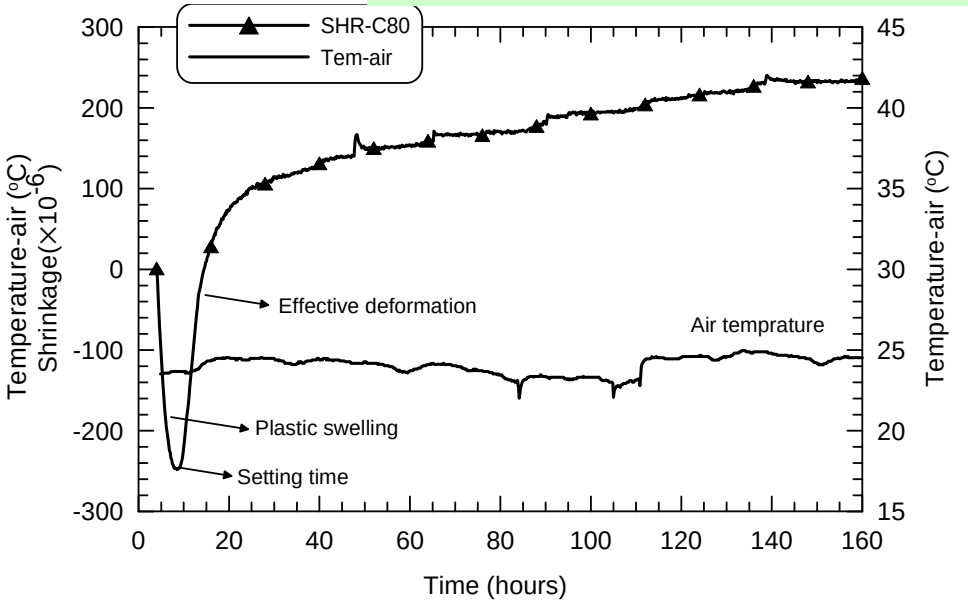


2. MECHANISMS





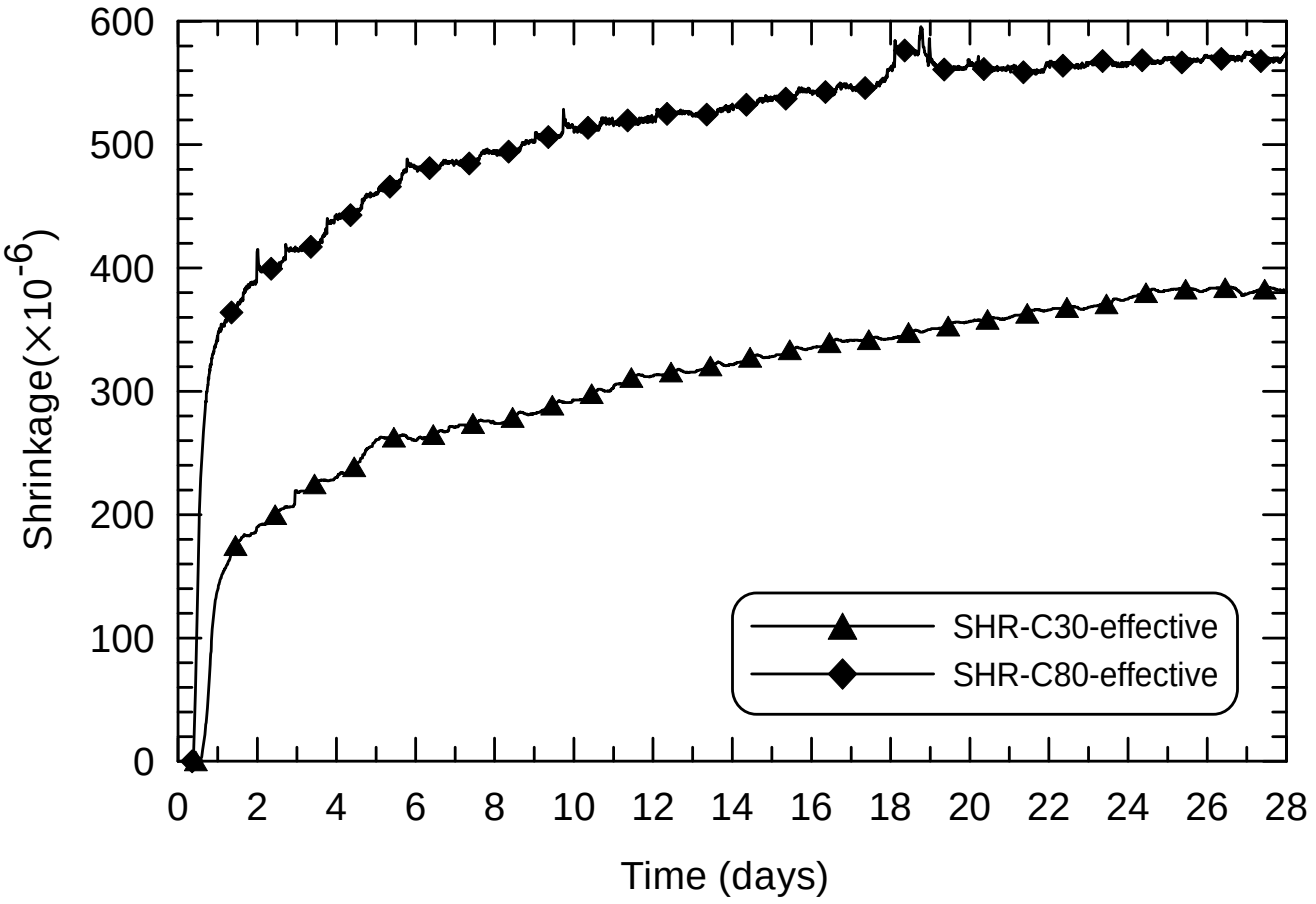




➢The global deformation first exhibits plastic swelling for the initial several hours after concrete casting and then shrinking. The end of swelling normally corresponds to the point of concrete setting, at which point the transformation of concrete from a liquid-like plastic state to a solid state is finished.

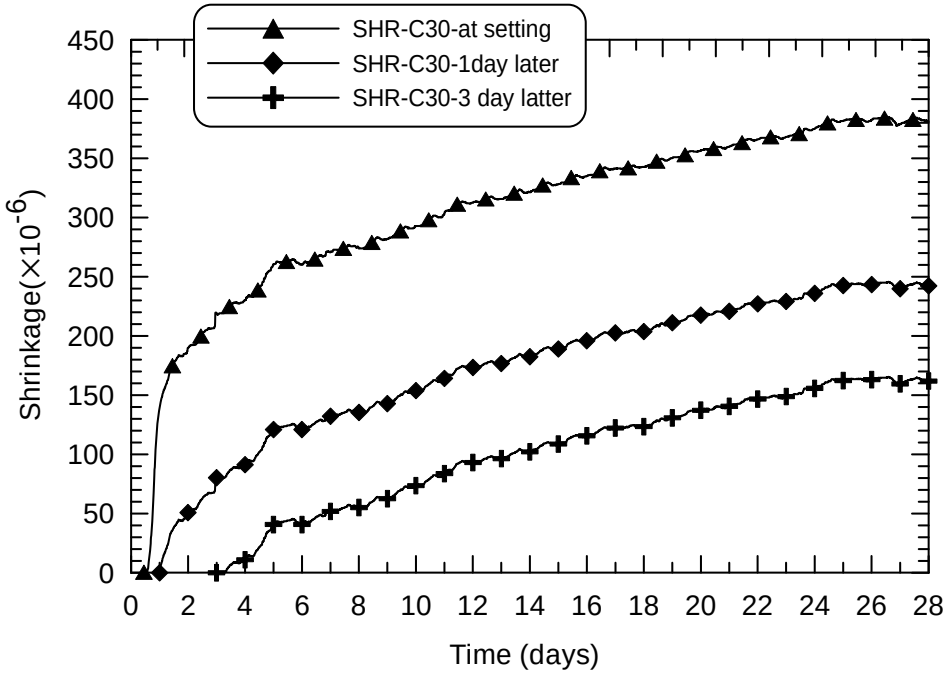
2. MECHANISMS

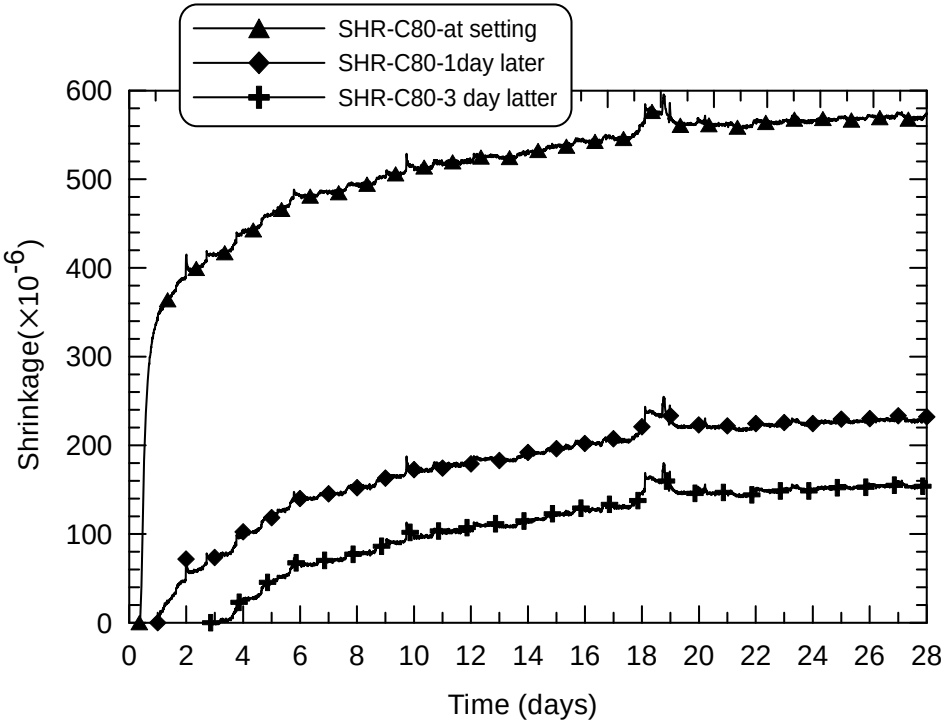
- **Complete shrinkage from setting**



2. MECHANISMS

❖ The effect of starting time on shrinkage of concrete

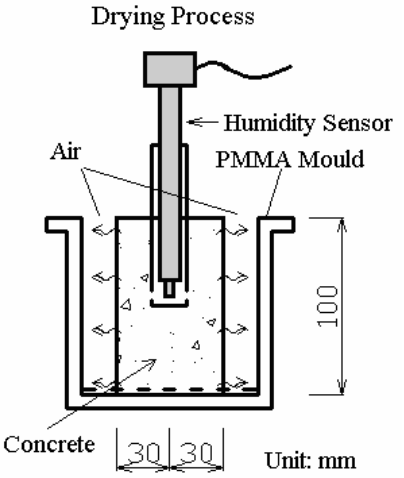


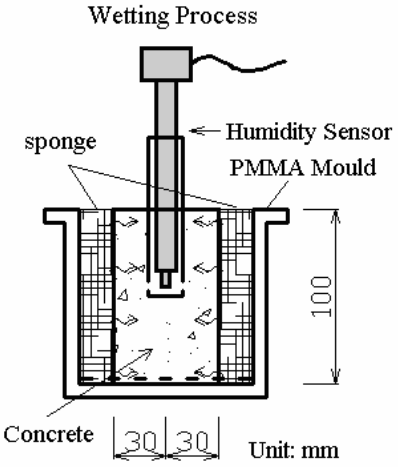


2. MECHANISMS

- **Integrative test on autogenous and drying shrinkages of concrete**







2. MECHANISMS

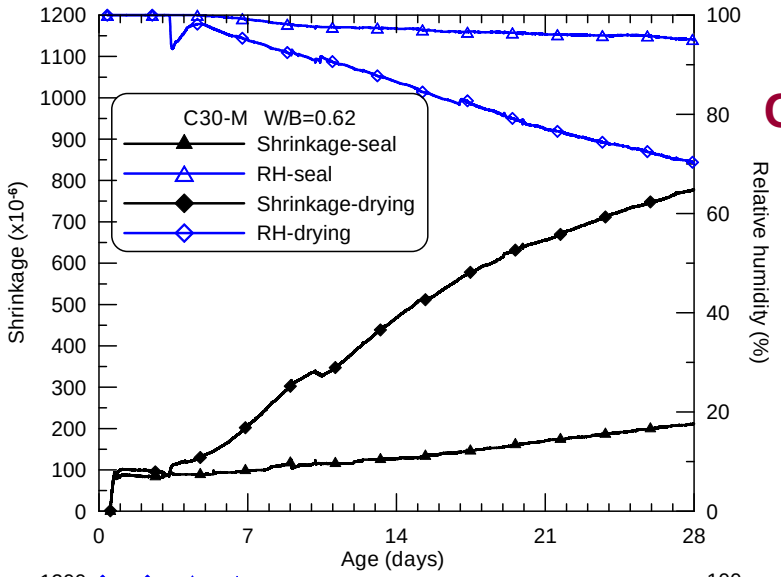
❖ Table 1. Mix proportions of concrete, kg/m^3

	No.	Cement	Water	Sand	Stone	Fly Ash	Silica Fume	*w/b*	V_a/V_c* (%)	Compressive strength (MPa)
C30 Series	C30-M	415	321	1296	0	104	-	0.62	49/0	38.8
	C30-V50	406	315	1268	58	102	-	0.62	50/2.1	27.9
	C30-V60	323	250	1010	591	81	-	0.62	60/21.9	28.0
	C30-V70	240	186	751	1125	60	-	0.62	70/41.7	30.9
C50 Series	C50- M	577	309	1146	0	142	-	0.43	44/0	62.0
	C50-V50	507	272	1006	325	125	-	0.43	50/12.0	54.3
	C50-V60	403	216	801	804	99	-	0.43	60/29.8	55.3
	C50-V70	300	161	596	1283	74	-	0.43	70/47.5	58.4
C80 Series	C80-M	831	277	1071	0	-	92	0.30	41/0	95.6
	C80-V50	694	231	895	438	-	77	0.30	50/16.2	75.5
	C80-V60	553	184	712	894	-	61	0.30	60/33.1	84.3
	C80-V70	411	137	530	1350	-	46	0.30	70/50.0	88.7

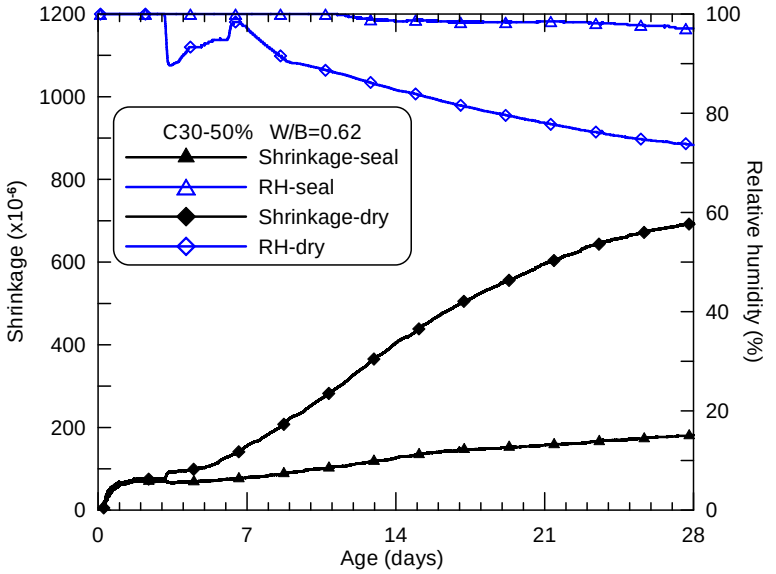
* V_a-total aggregate content in volume, V_c-coarse aggregate content in volume

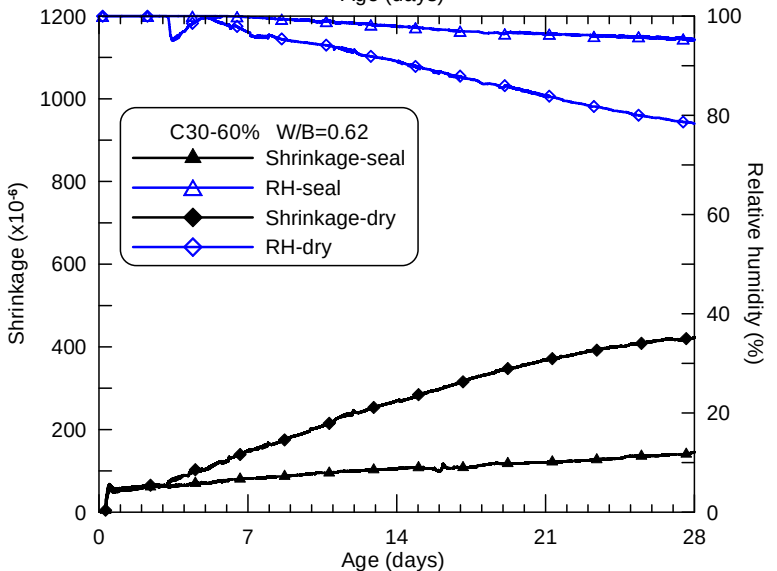
2. MECHANISMS

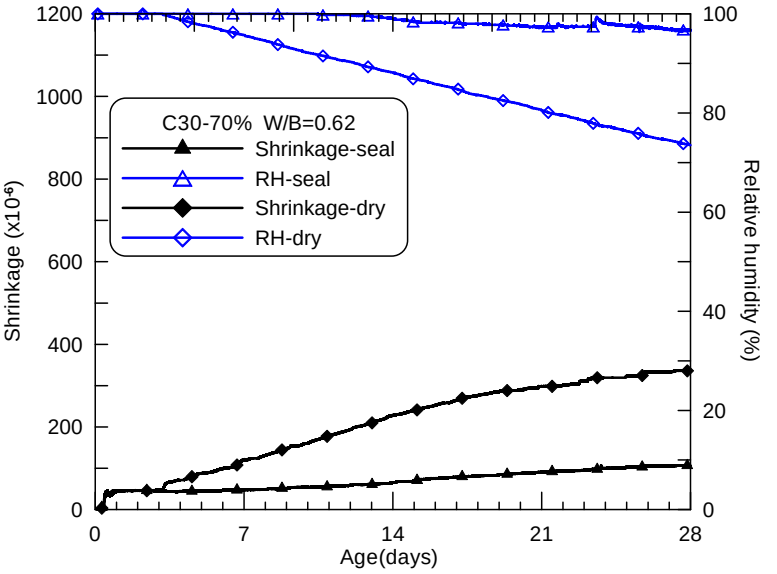
Experimental Observations



C30 Series

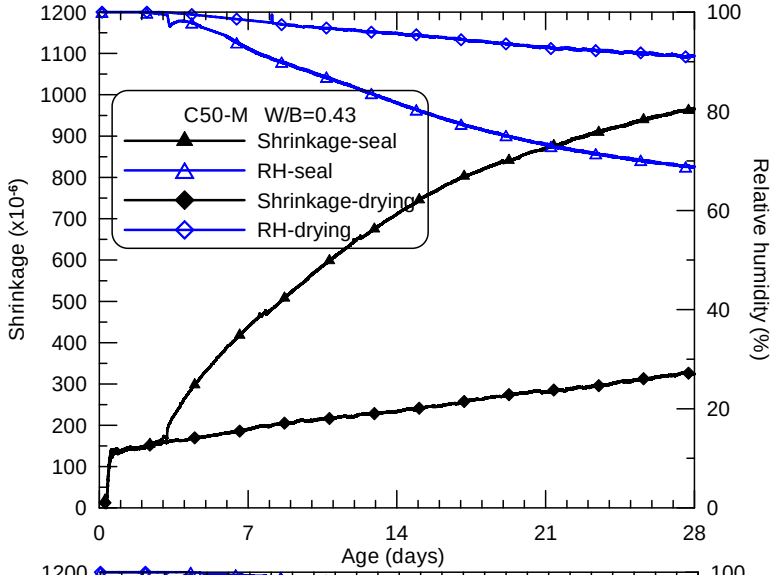




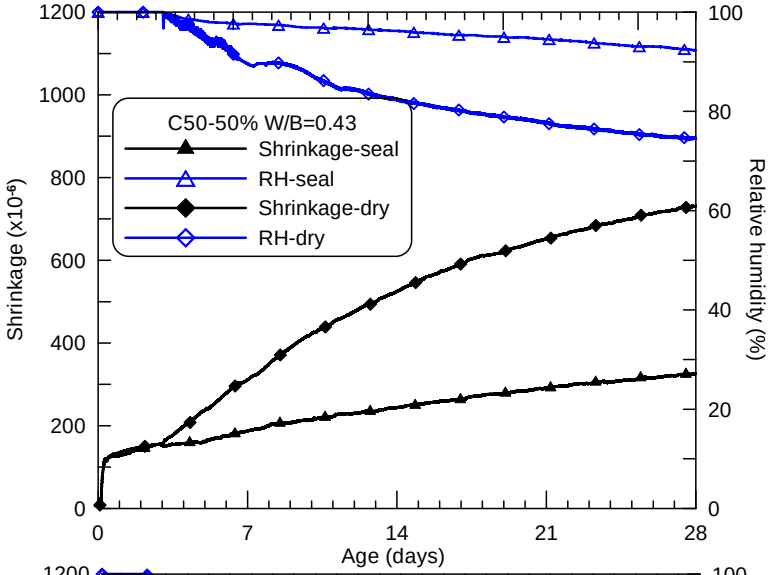


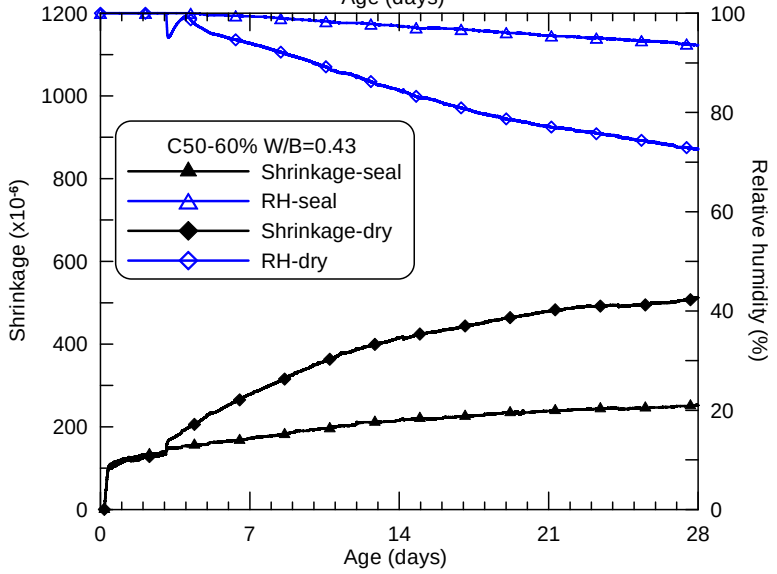
2. MECHANISMS

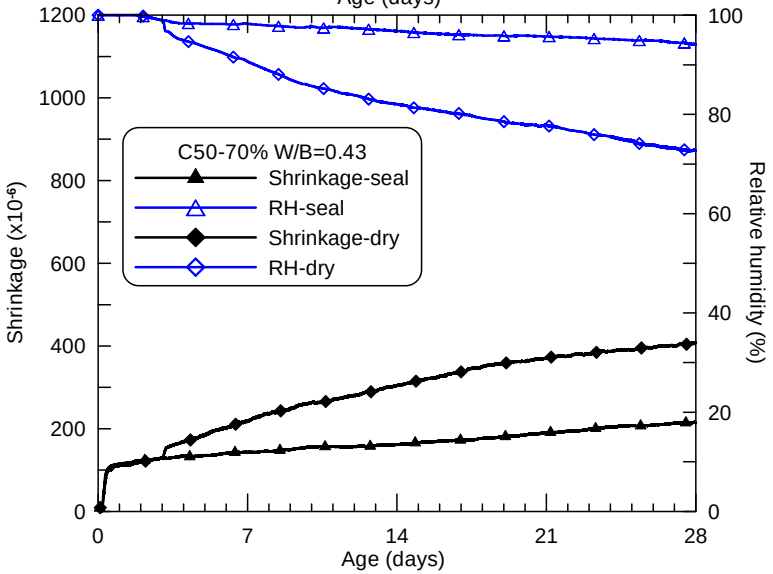
❖ Experimental Observations



C50 Series

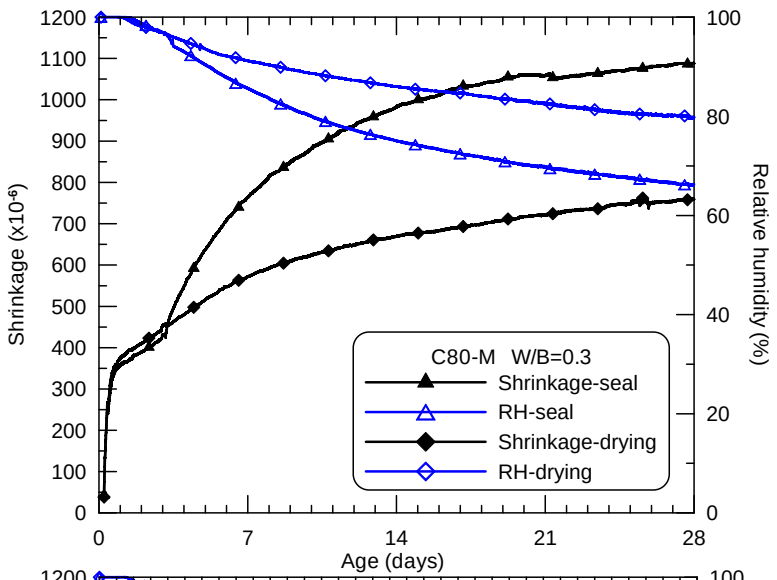




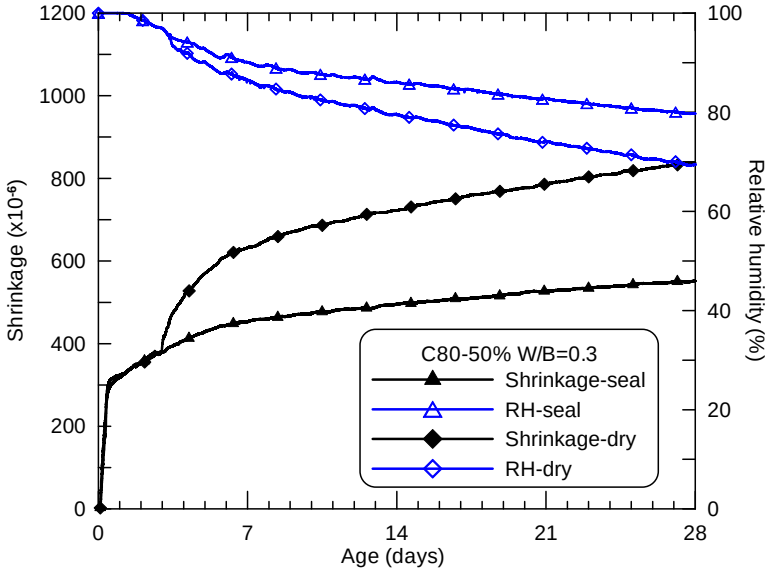


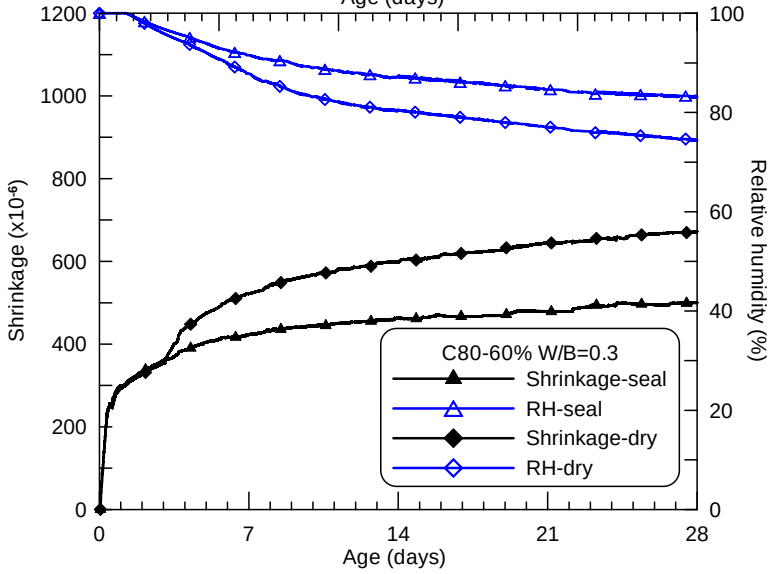
2. MECHANISMS

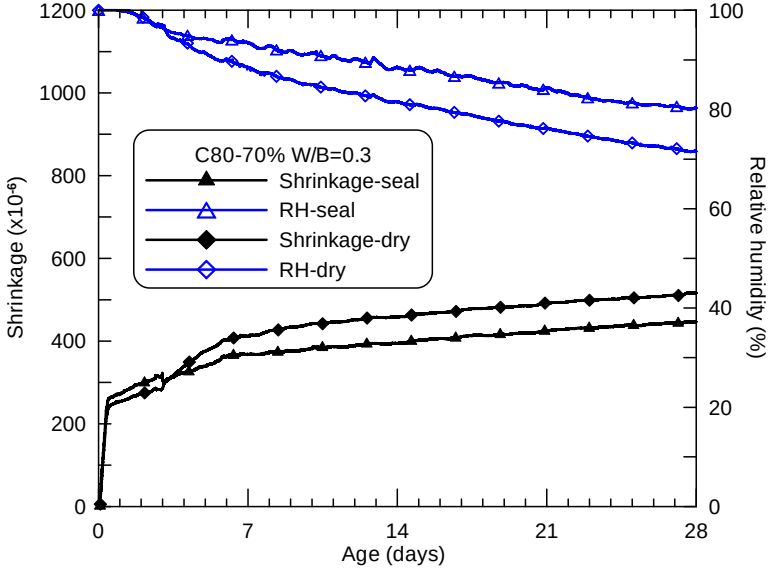
Experimental Observations



C80 Series

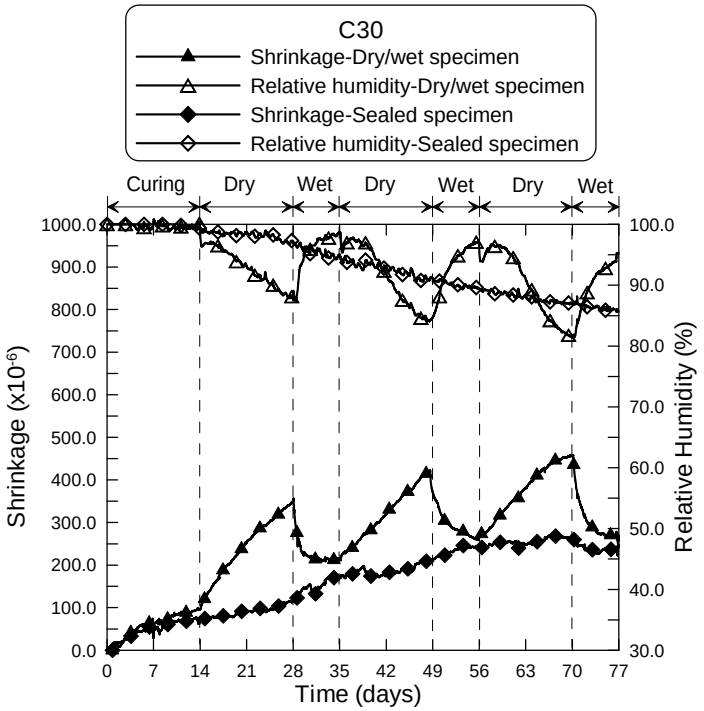


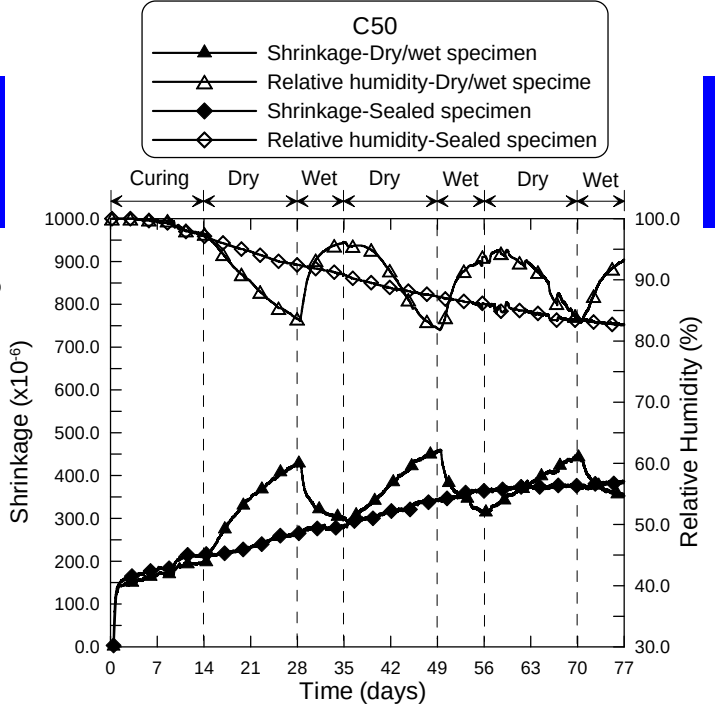


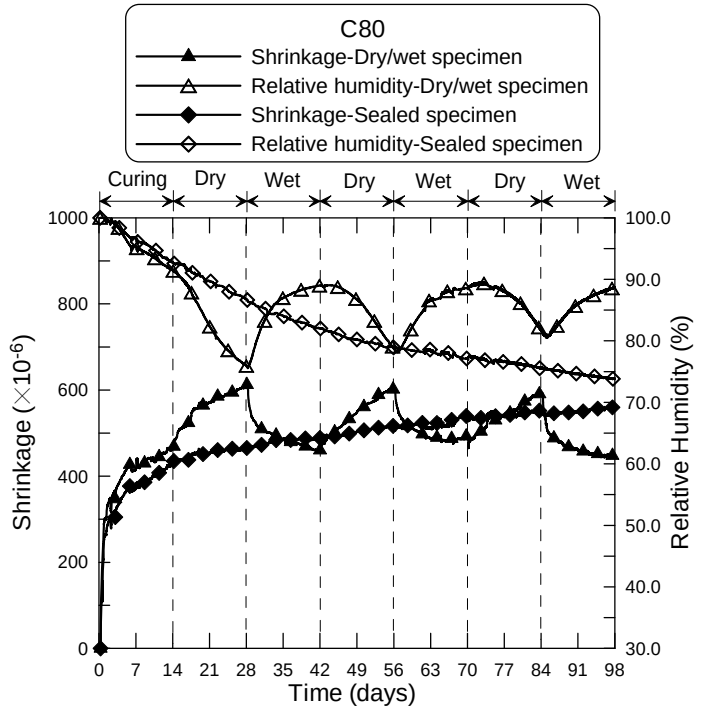


2. MECHANISMS

❖ Experimental Observations (under dry-wet cycles)

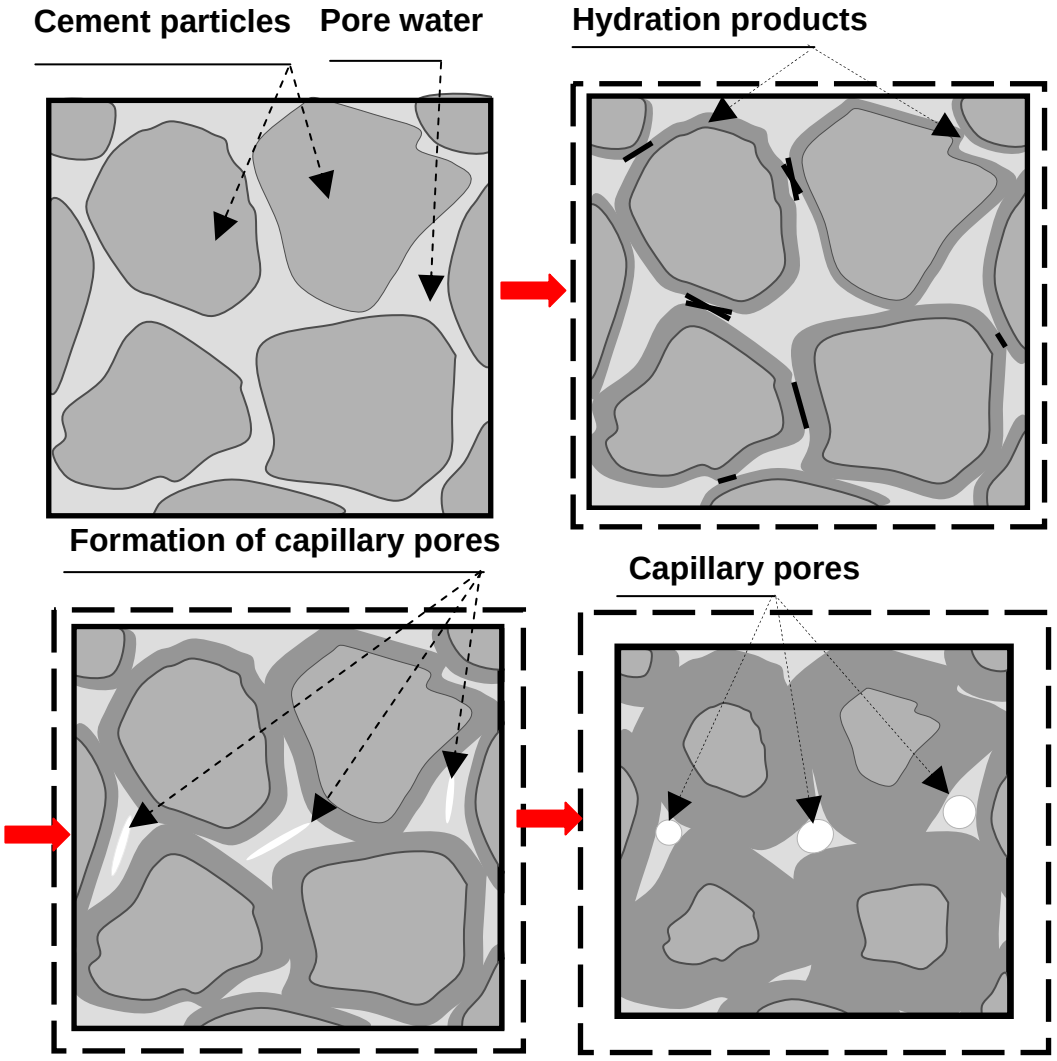


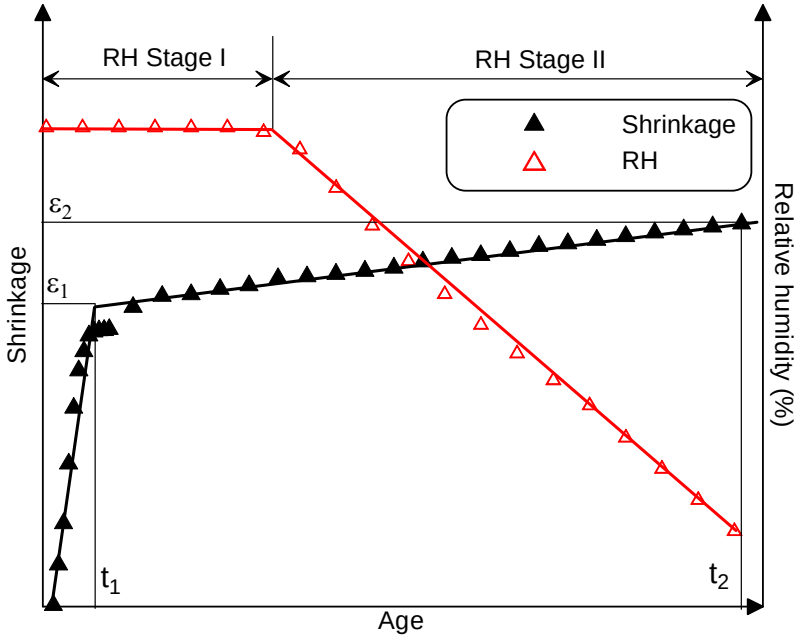




2. MECHANISMS

Mechanism analyses





- At an initial short period after cast, a continuous liquid network is existed and that results in the relative humidity inside of concrete is 100%.
- As the connection area between solid particles is sufficient to support the self-weight of concrete, setting of concrete occurs.
- The continuity of the pore water is finally broken down and the pore water is isolated into some individual pocket lacks without direct connection between each others, RH<100%.

2. MECHANISMS

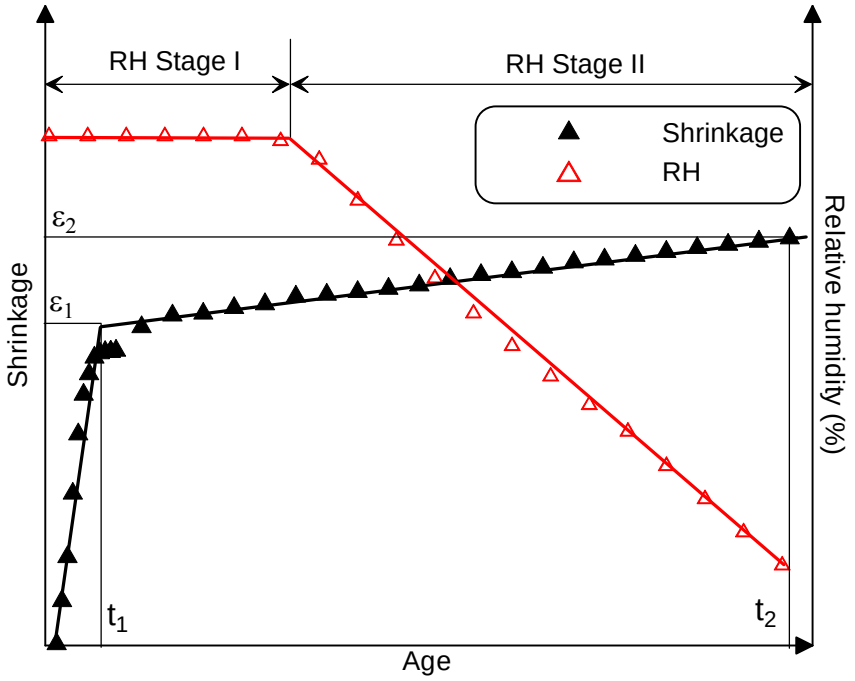
Chemical shrinkage

Water evaporation

ΔV

$$\varepsilon_w = k\Delta V \qquad \Delta V = f(\alpha)$$

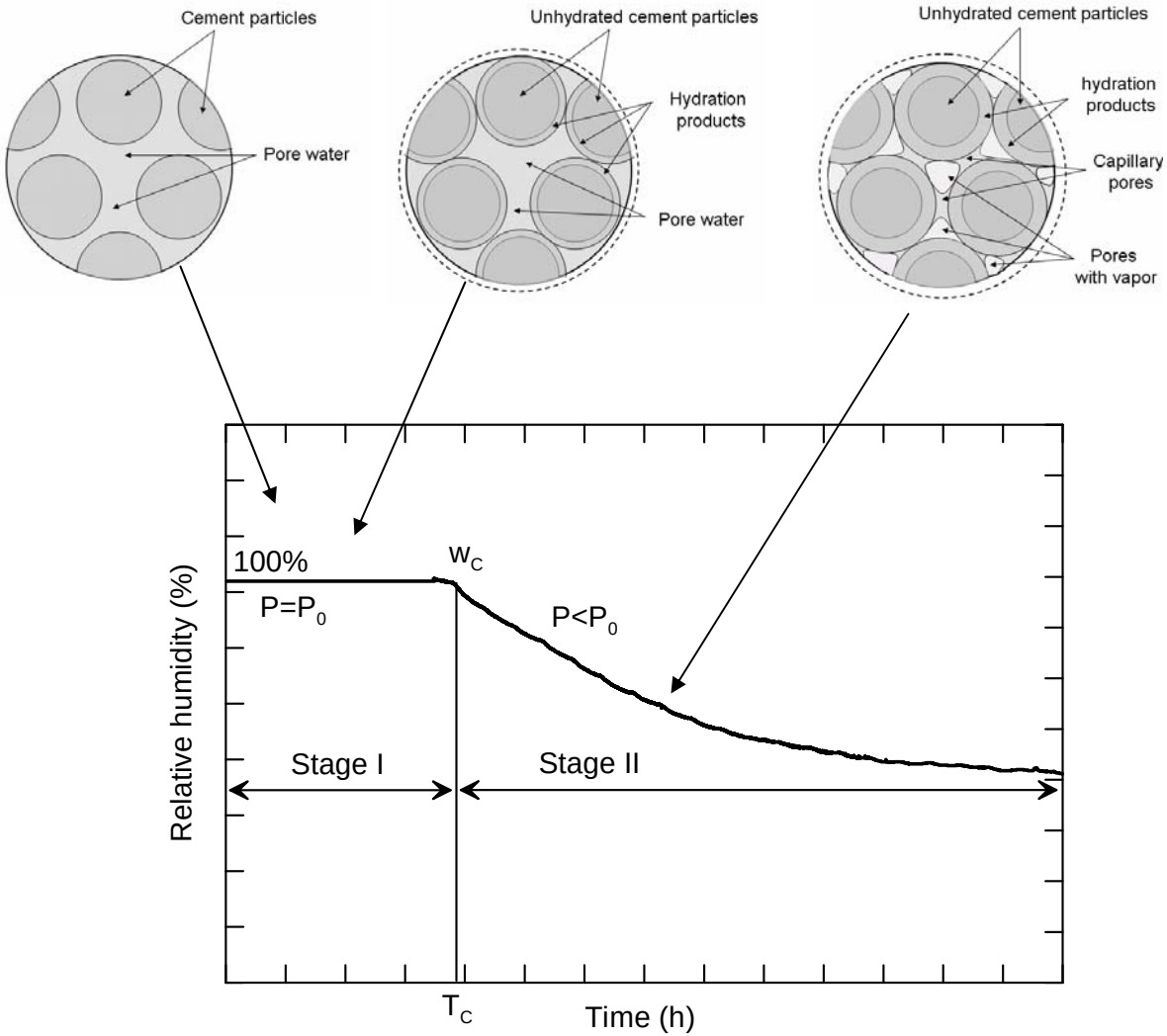
$$\sigma_w = -\frac{2\gamma_{LV}}{r_c} \qquad \sigma_w = -\frac{\rho_w R_g T}{M}\ln(RH) \qquad \varepsilon_w = f(\sigma_w)$$



- Test results show that the development of internal relative humidity since concrete casting obeys a two-stage mode;
- Chemical shrinkage and capillary tension generated in moisture saturated stage and humidity reduction stage should be the main resource leading concrete to shrink in each stage respectively.
- The shrinkage of concrete since set may be simulated as well individually according to the stages of internal relative humidity progress in concrete.

3. SIMULATION

- **Modeling on autogenous and drying shrinkages of concrete**



$$t_e = \int_0^t e^{\frac{1}{R}\left(\frac{U_{ar}}{293} - \frac{U_{aT}}{273+T}\right)} dt$$

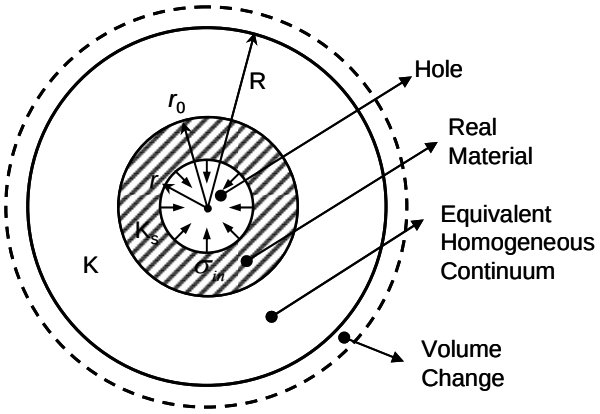
$$\alpha = \alpha_u \exp - \left(\frac{A}{t_e}\right)^B$$

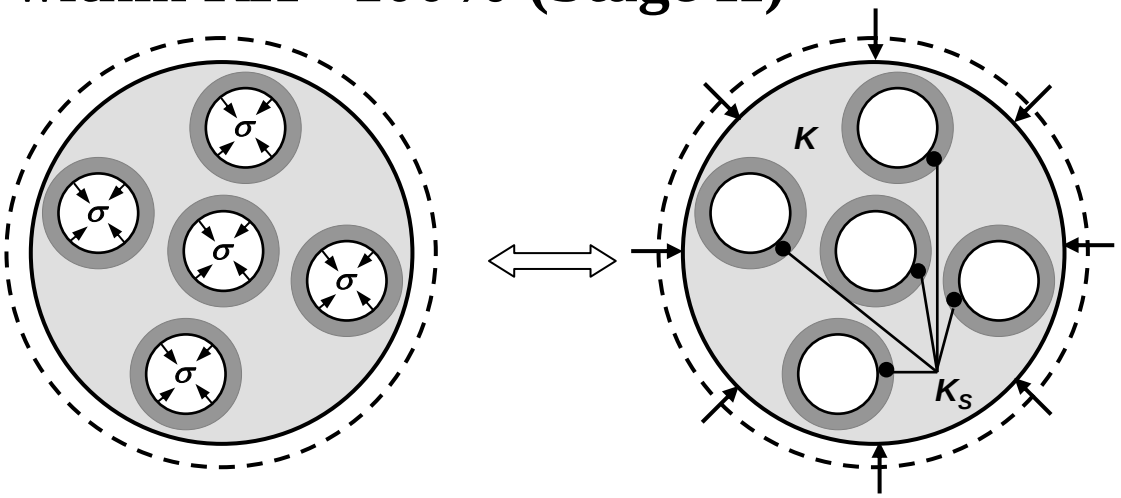
$$\alpha_u = \frac{1.031 \, w/c}{0.194 + w/c}$$

$$E = 1.05 E_{28} \left(\frac{a - a_c}{\alpha_u - a_c}\right)^{1/2}$$

3. SIMULATION

- **Shrinkage developed within RH <100% (Stage II)**

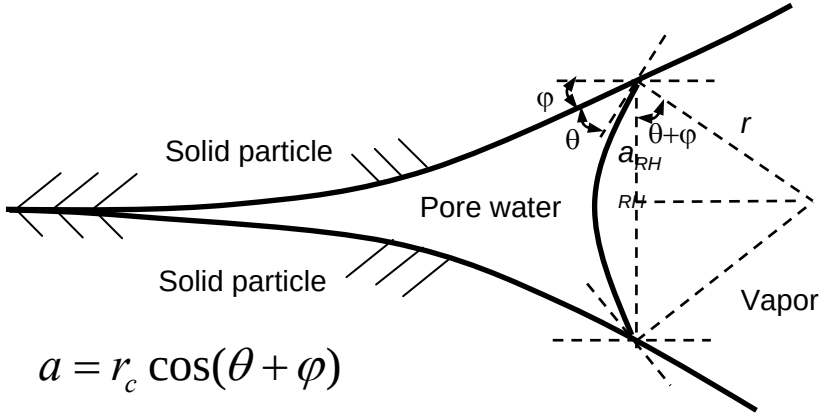




$$\varepsilon_w = \frac{\sigma_w}{3}\left(\frac{1}{K}-\frac{1}{K_s}\right)\left(\frac{r_0}{R}\right)^3$$

$$\varepsilon_w = \sum_{i=1}^{N}\frac{\sigma_{wi}}{3}\left(\frac{1}{K}-\frac{1}{K_s}\right)\left(\frac{r_0}{R_i}\right)^3 = \frac{\sigma_w}{3}\left(\frac{1}{K}-\frac{1}{K_s}\right)$$

$$\varepsilon_w = \frac{\sigma_w}{3}\left(\frac{1}{K}-\frac{1}{K_s}\right)\int_0^{a_{RH}} s(a)da$$



$$a = r_c\cos(\theta+\varphi)$$

$$\varepsilon_w = -\nu_p\frac{\rho_w R_g T}{3M}\left(\frac{1}{K}-\frac{1}{K_s}\right)\ln(RH)$$

$$\nu_p = \int_0^{a_{RH}} s(a)da = 1-\exp(k_0\beta a_{RH})$$

3. SIMULATION

❖ Shrinkage developed within RH =100% (Stage I)

The chemical shrinkage can be estimated by Powers' volumetric models:

$$V_{cs} = 0.2(1-p)\alpha \qquad V_{cs} = k(0.2+0.7s/c)(1-p)\alpha$$

$$p = \frac{w/c}{w/c + \rho_w/\rho_c} \qquad p = \frac{w/c}{w/c + \rho_w/\rho_c + (\rho_w/\rho_c)(s/c)}, \qquad k = \frac{1}{1+1.4(s/c)}$$

$$\varepsilon_w = \frac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs}-V_{cs0})}\right]$$

V_{cs0} - the chemical shrinkage at concrete set.

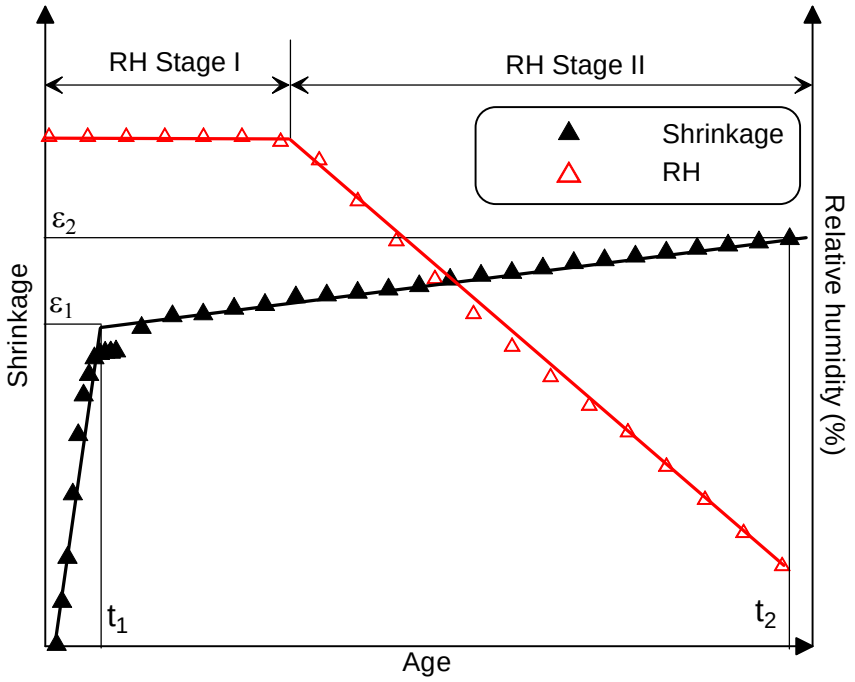
ϕ_P - volume fraction of cement paste.

K_0, η - constants.

3. SIMULATION

❖ Complete shrinkage model

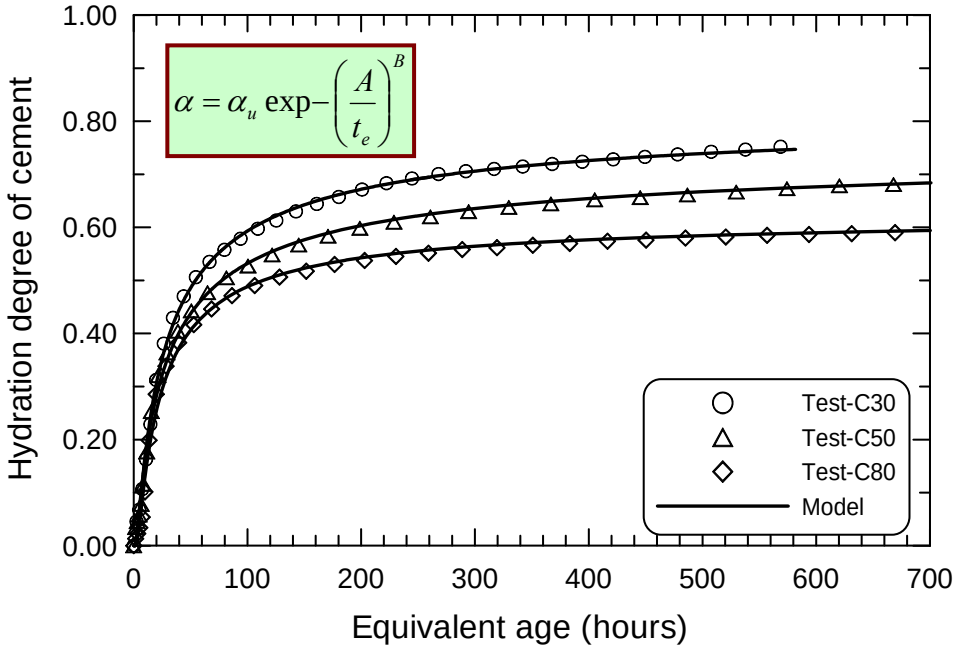
$$\varepsilon_w = \begin{cases} \dfrac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs}-V_{cs0})}\right] & \text{for RH} = 1 \\ \dfrac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs1}-V_{cs0})}\right]+\dfrac{\nu_p \rho_w R_g T}{3M}\left(\dfrac{1}{K_s}-\dfrac{1}{K}\right)\ln(RH) & \text{for RH} < 1 \end{cases}$$

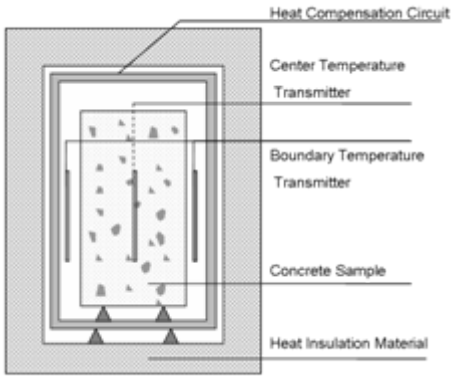


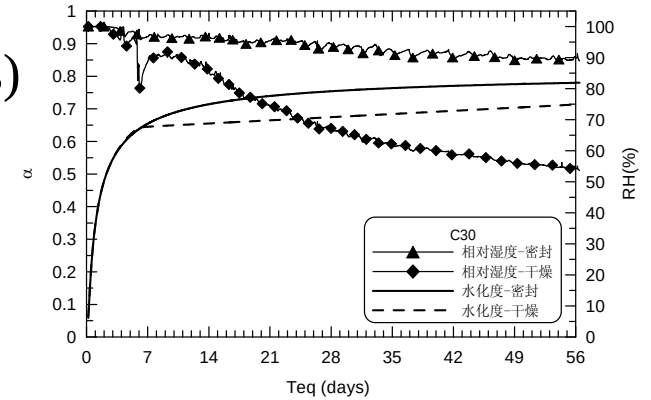
- The model can be used for water loss resulting shrinkage prediction regardless if the water loss is caused by cement hydration or by environmental drying.
- By applying the model, calculation of shrinkage distribution in concrete structures in the case of humidity gradient existing becomes possible.
- The influences of curing conditions on concrete shrinkage should be able to be reflect in the model by cement hydration degree and internal relative humidity.

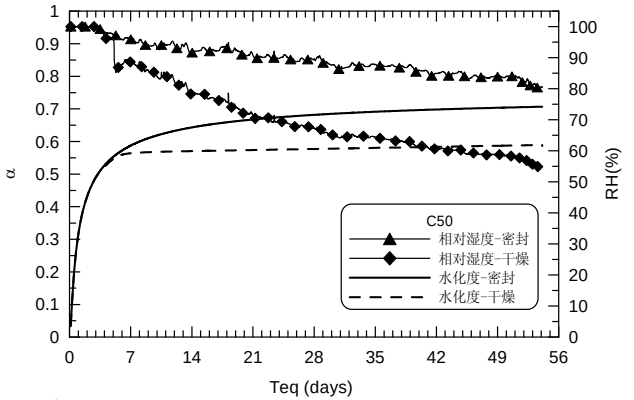
3. SIMULATION

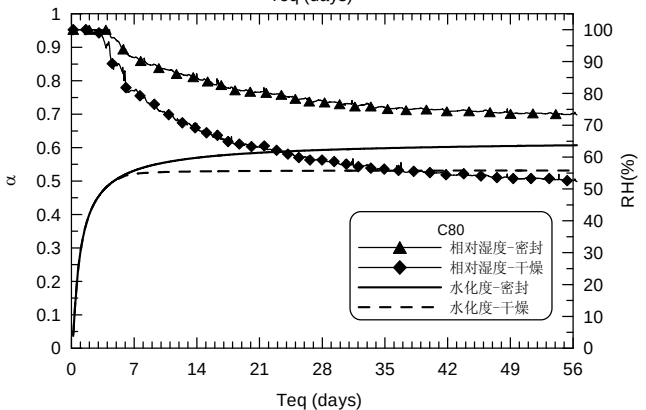
- Cement hydration model (isothermal tests)





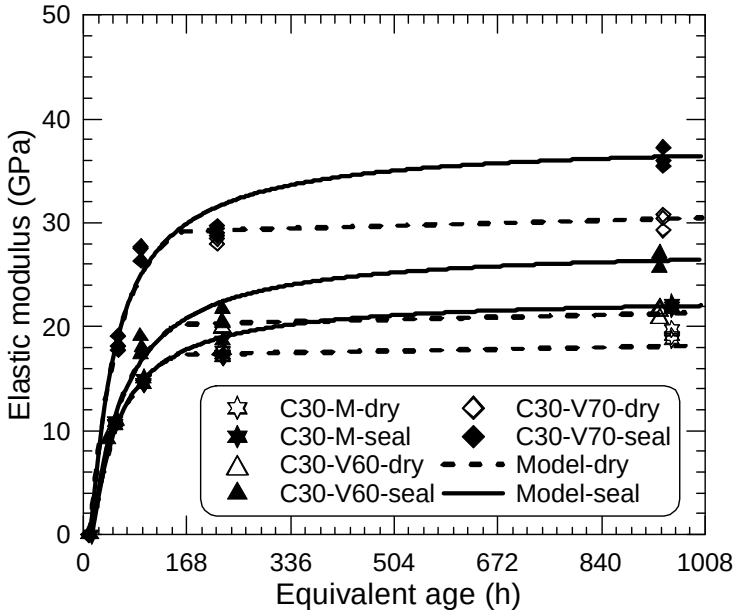




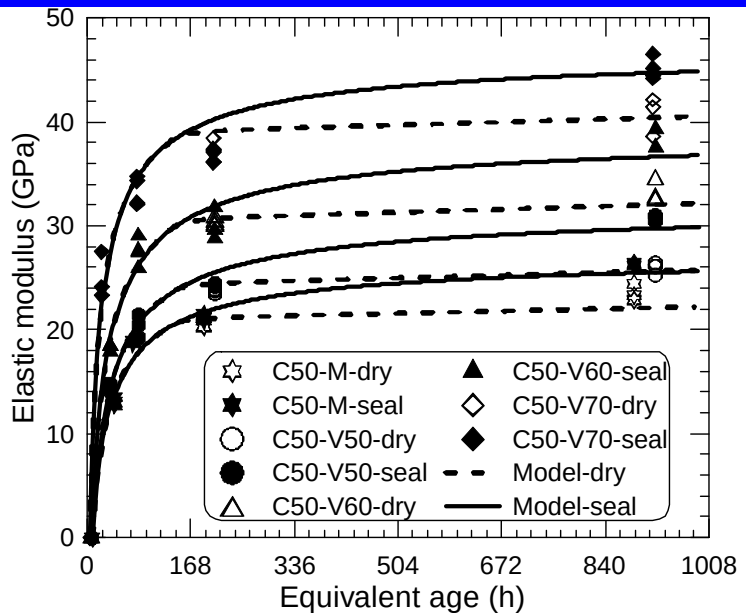


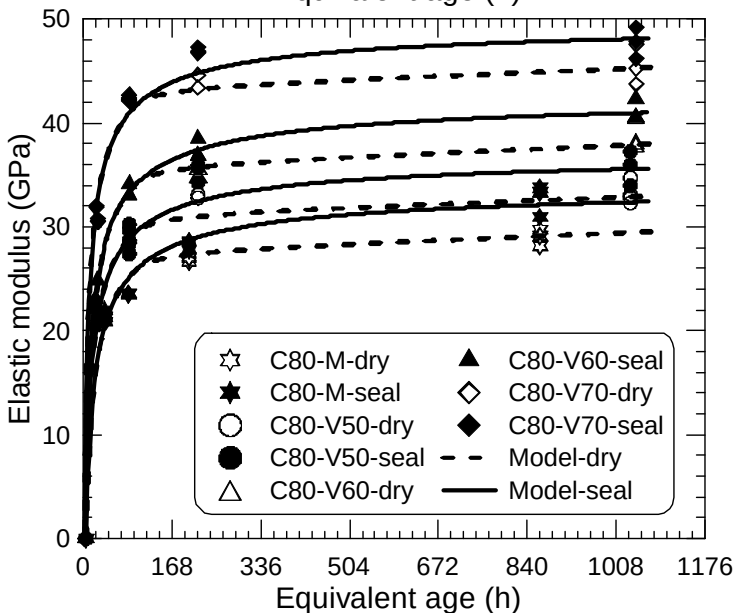
3. SIMULATION

❖ Elastic modulus



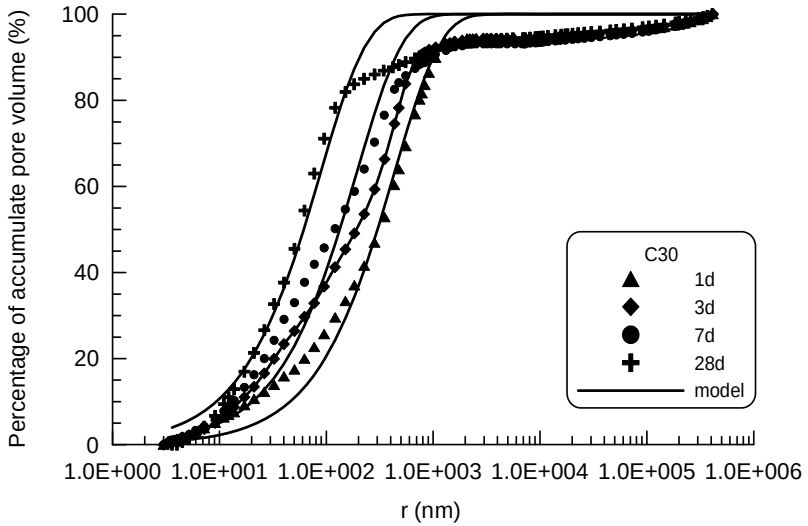
$$E = 1.05E_{28}\left(\frac{a-a_c}{\alpha_u-a_c}\right)^b$$

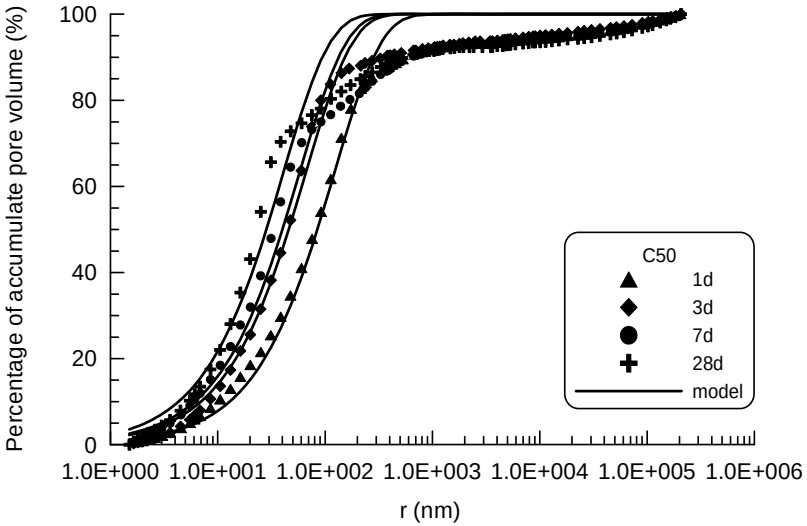


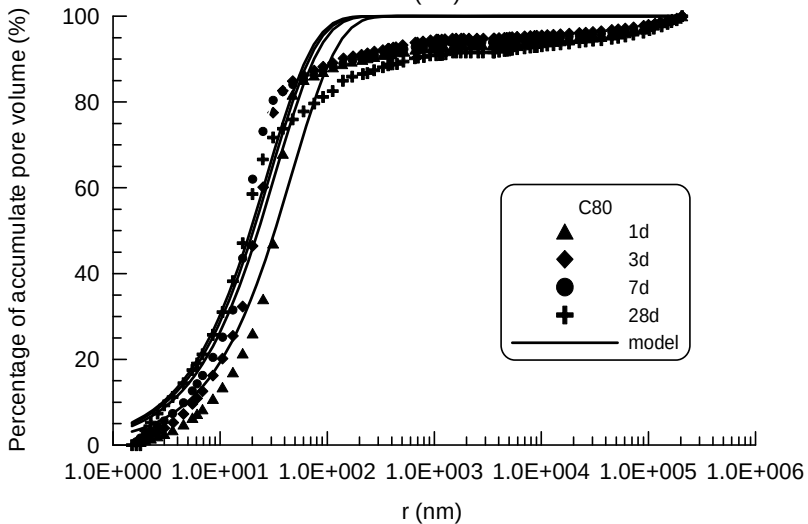


3. SIMULATION

- **Pore size influencing factor ν_p** (mercury penetration experiments)







$$\nu_p = \int_0^{a_{RH}} s(a)da = 1 - \exp(k_0 \beta a_{RH})$$

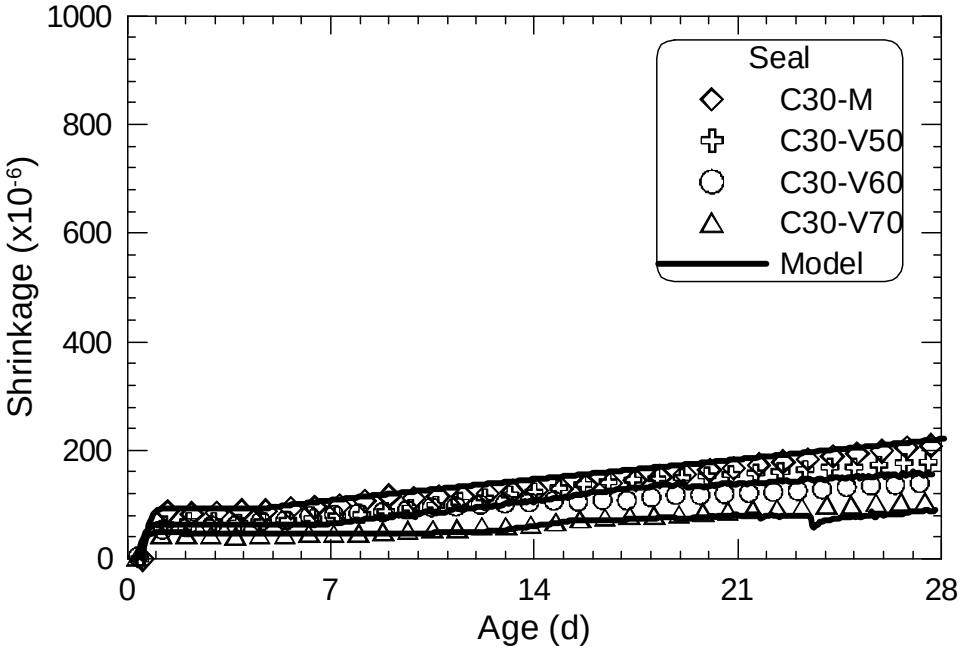
3. SIMULATION

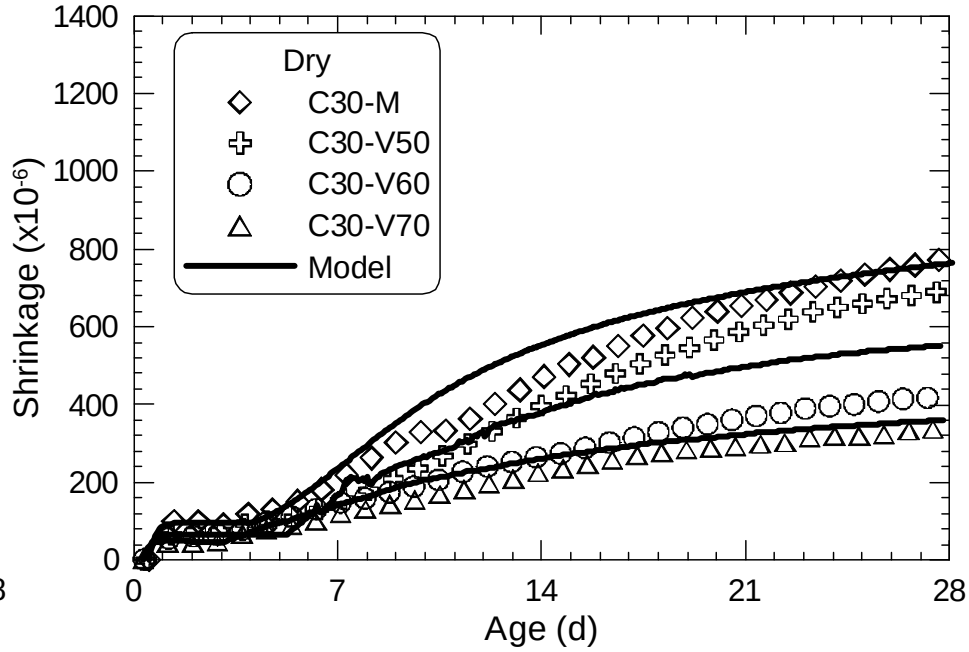
❖ Table 2 Parameters used in shrinkage calculation

No.	ϕ_p	Stiffness constants		Hydration degree parameter					Elastic modulus parameter		Pore structure parameter			
		K_0	η	α_u	A	B	α_0	α_c	E_{28} (GPa)	E_s (GPa)	b	k_0	μ	λ
C30-M	0.455	5.0	1.60	0.885	32.600	0.9604	0.079	0.615	22.0	79.2	1.13	17.15	0.0007	4.375
C30-V50	0.446		1.60				0.077		22.0		1.13			
C30-V60	0.355		1.59				0.028		26.5		1.35			
C30-V70	0.264		1.43				0.021		36.3		1.14			
C50- M	0.495	4.5	1.34	0.809	20.182	0.6936	0.134	0.528	26.3		0.93	25.58	0.0030	3.057
C50-V50	0.435		1.27				0.112		30.8		1.01			
C50-V60	0.346		1.16				0.088		37.6		0.95			
C50-V70	0.258		1.04				0.051		45.1		0.76			
C80- M	0.587	4.5	0.96	0.626	13.323	0.6281	0.125	0.375	32.9		0.79	64.29	0.0112	2.224
C80-V50	0.490		0.94				0.103		35.7		0.69			
C80-V60	0.390		0.84				0.089		41.1		0.70			
C80-V70	0.290		0.76				0.057		47.8		0.59			

3. SIMULATION

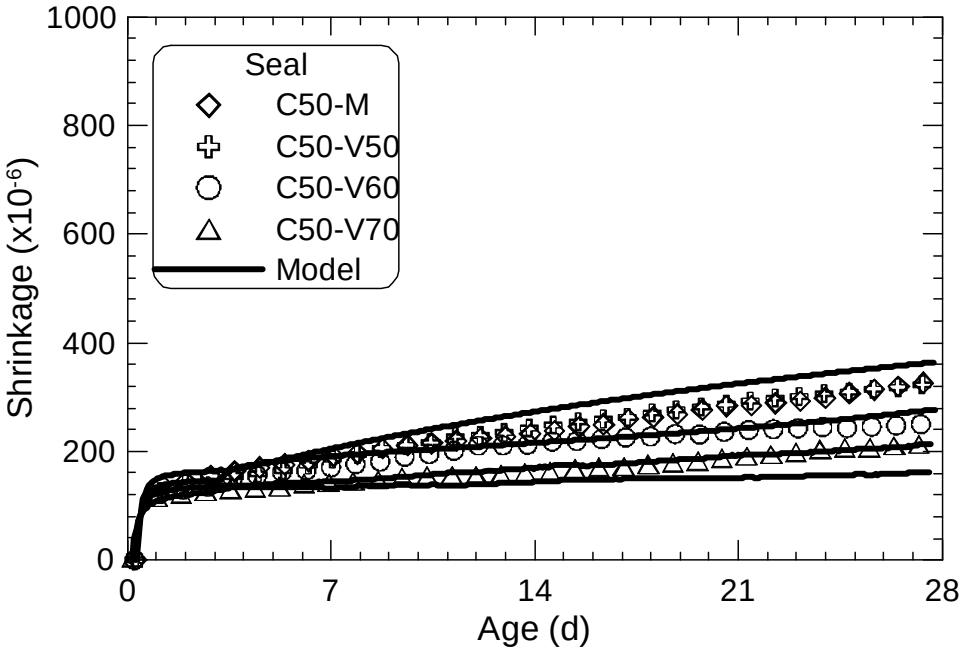
❖ Model results (C30 Series)

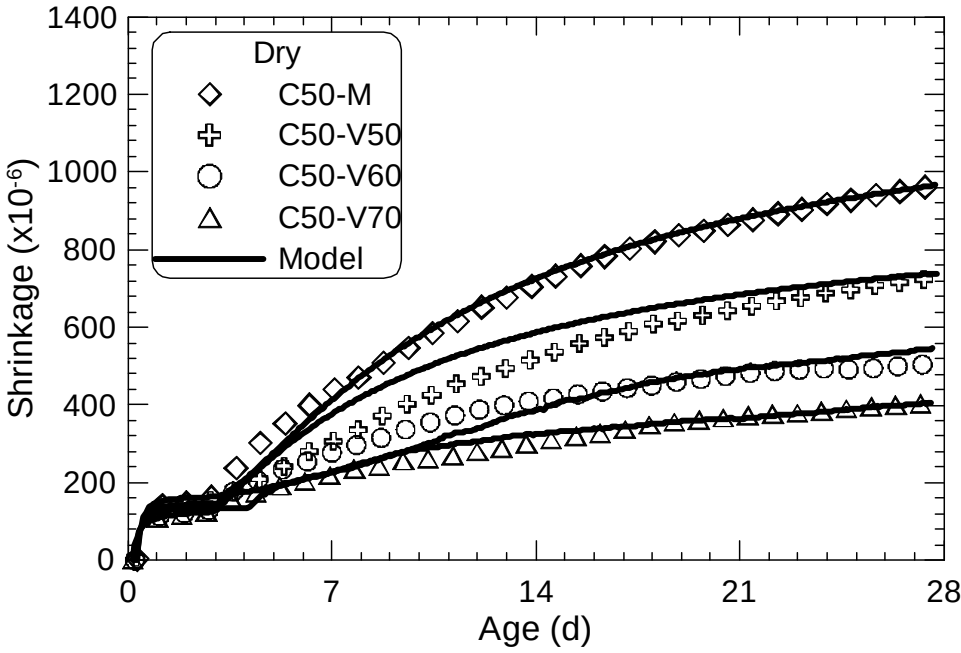




3. SIMULATION

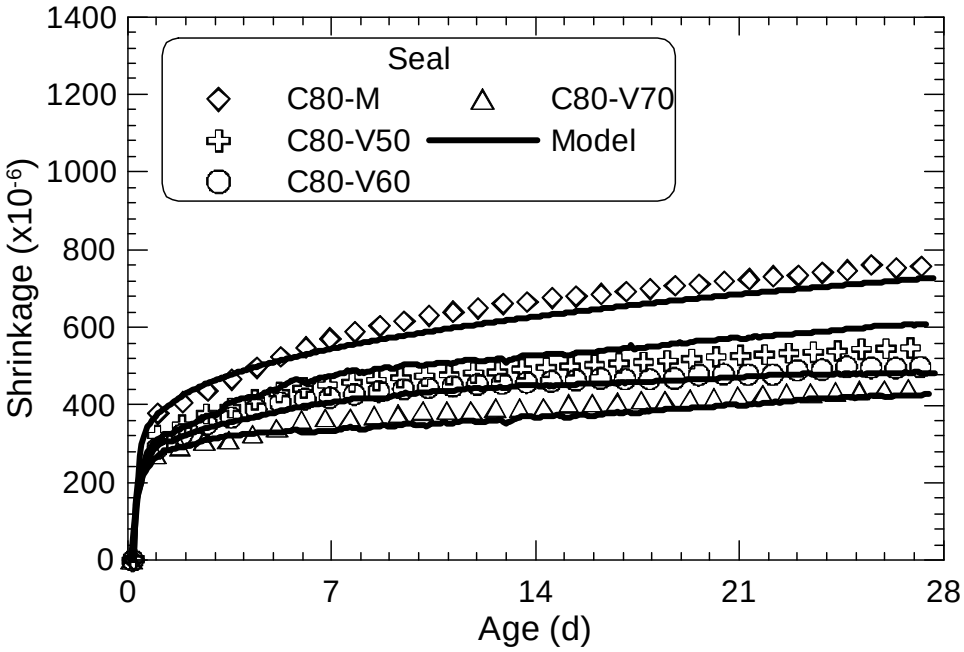
❖ Model results (C50 Series)

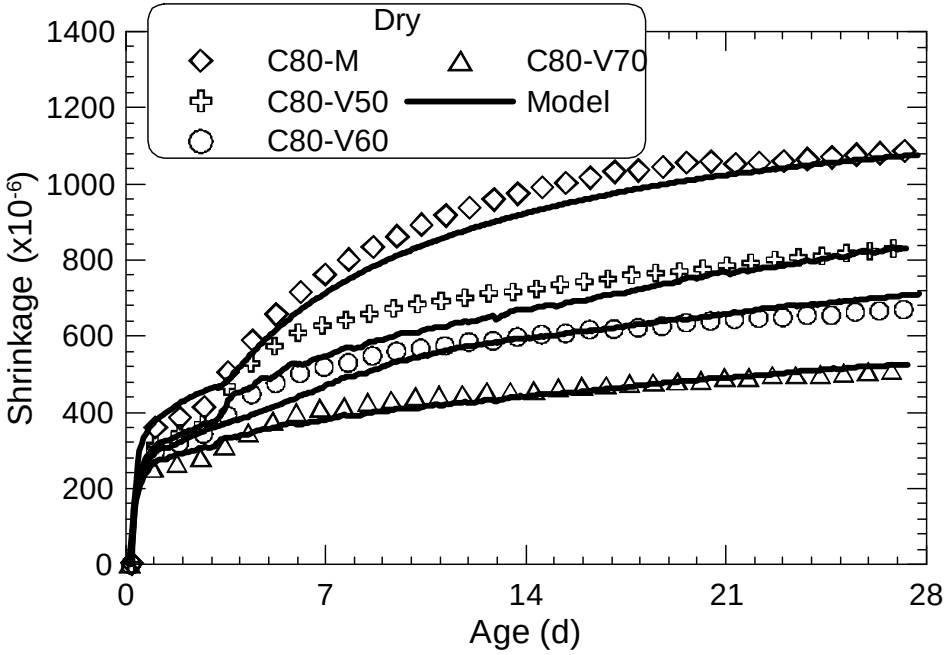




3. SIMULATION

❖ Model results (C80 Series)

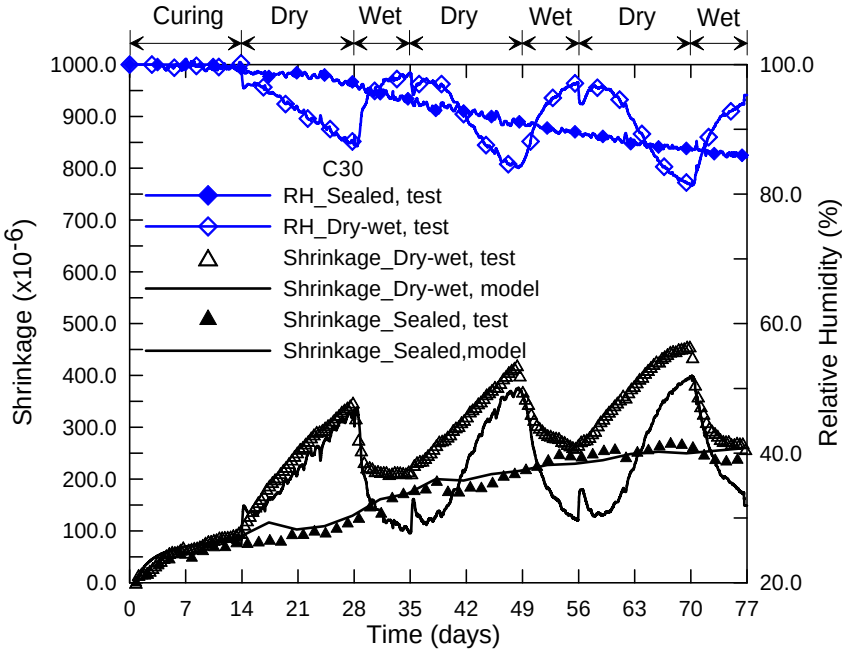


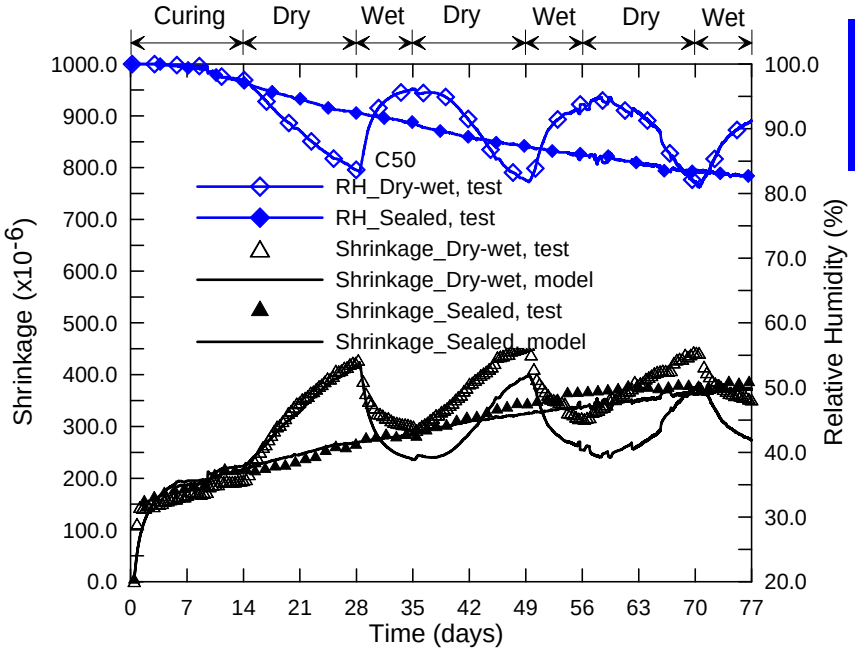


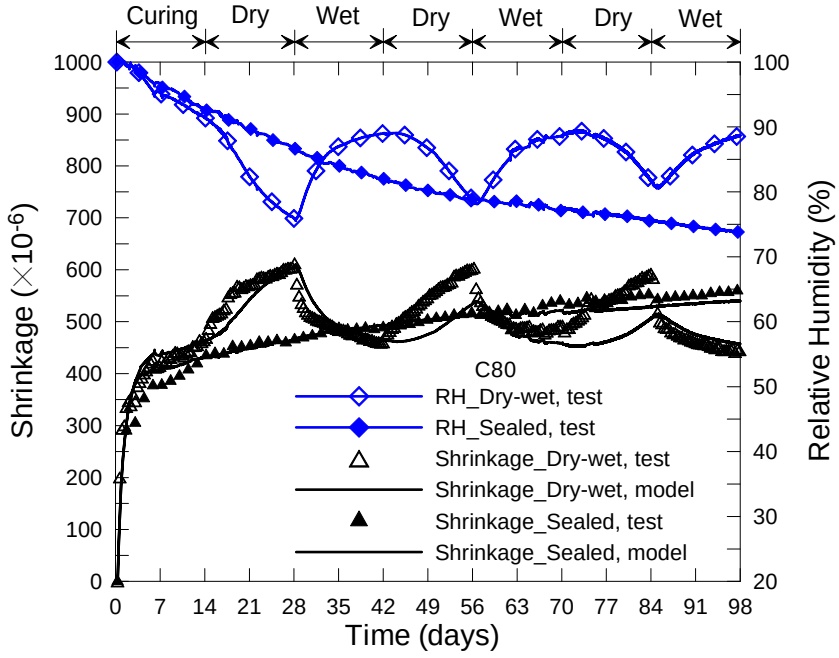
3. SIMULATION

❖ **Model results**

(under dry-wet cycles)







4. REGULATION

$$\varepsilon_w = \begin{cases} \dfrac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs}-V_{cs0})}\right] & \text{for RH}=1 \\ \dfrac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs1}-V_{cs0})}\right]+\dfrac{\nu_p \rho_w R_g T}{3M}\left(\dfrac{1}{K_s}-\dfrac{1}{K}\right)\ln(RH) & \text{for RH}<1 \end{cases}$$

- **To reduce shrinkage of concrete, two technical strategies may normally be considered.**
- **One is to reduce the driving force, such as using internal curing and/or using specific surfactants in water.**
- **Another way is to enhance the shrinkage resistance, such as to increase the elastic modulus of concrete, typically using more aggregates as present in the previous section.**

?

$$\sigma_w = -\frac{2\gamma_{LV}}{r_c}$$

K

?

4. REGULATION

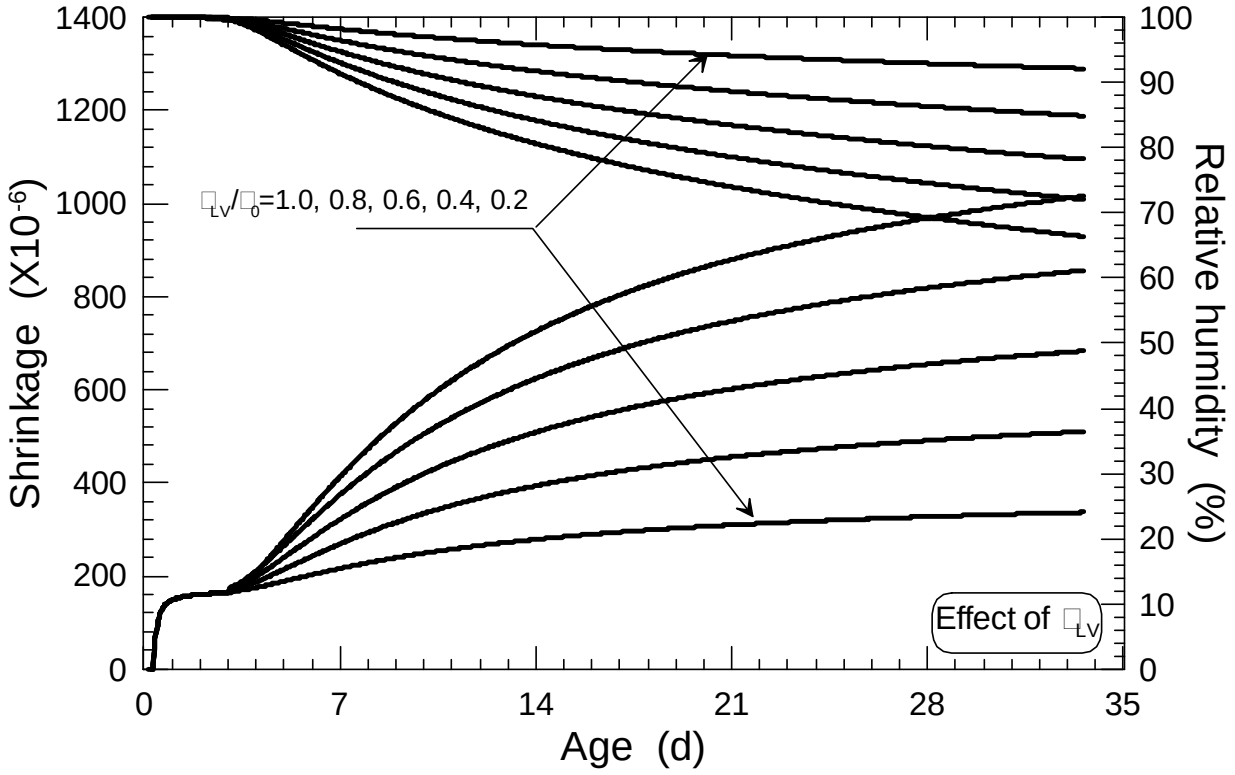
❖ Effect of surface tension of water

$$\varepsilon_w = \begin{cases} \dfrac{\phi_P}{(K+K_0)^\eta}\left[1-\sqrt[3]{1-(V_{cs}-V_{cs0})}\right] & \text{for RH} = 1 \\ \dfrac{\phi_P}{(K+K_0)^\eta}\left[1-\sqrt[3]{1-(V_{cs1}-V_{cs0})}\right]+\dfrac{\nu_p \rho_w R_g T}{3M}\left(\dfrac{1}{K_s}-\dfrac{1}{K}\right)\ln(RH) & \text{for RH} < 1 \end{cases}$$

$$\sigma_w = \frac{2\gamma_{LV}}{r_c} = -\frac{\rho_w R_g T}{M}\ln(RH)$$

$$\gamma_{cw} = \phi\gamma_w$$

$$\varepsilon_{cw} = \phi\varepsilon_w$$



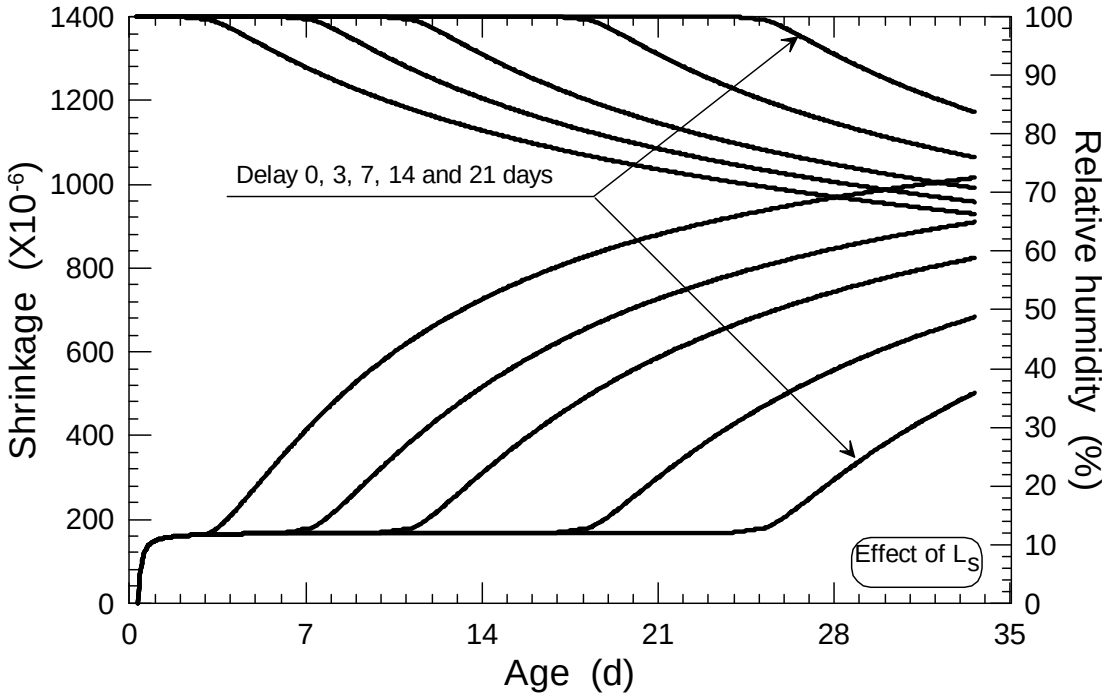
4. REGULATION

❖ Effect of curvature of capillary pores

$$\varepsilon_w = \begin{cases} \dfrac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs}-V_{cs0})}\right] & \text{for RH} = 1 \\ \dfrac{\phi_P}{(K+K_0)^{\eta}}\left[1-\sqrt[3]{1-(V_{cs1}-V_{cs0})}\right]+\dfrac{\nu_p \rho_w R_g T}{3M}\left(\dfrac{1}{K_s}-\dfrac{1}{K}\right)\ln(RH) & \text{for RH} < 1 \end{cases}$$

$$\sigma_w = \frac{2\gamma_{LV}}{r_c} = -\frac{\rho_w R_g T}{M}\ln(RH)$$

➢Apart from reducing surface tension of water, another way to decrease driving force of shrinkage is to refill the capillary pores by internal water sources, for example using pre-saturated lightweight aggregates or super-absorbent polymer in concrete.



4. REGULATION

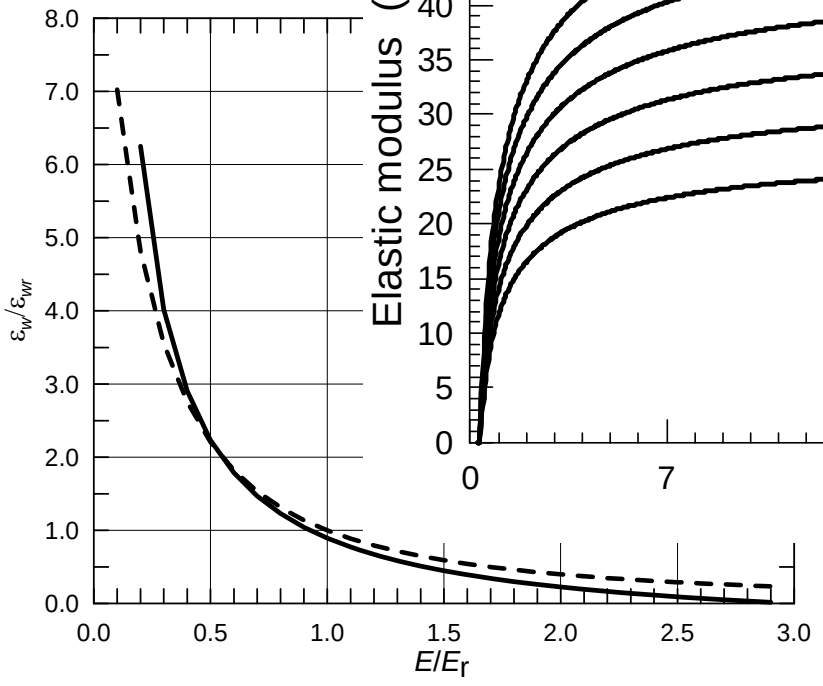
❖ Effect of elastic modulus of concrete

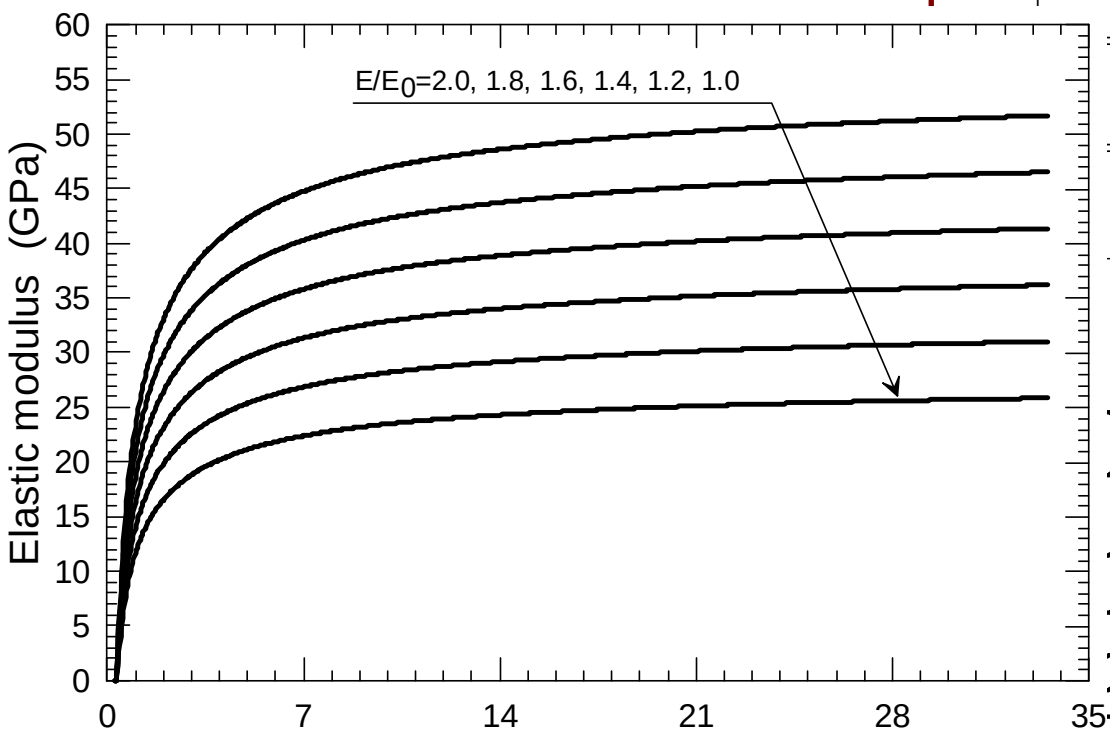
$$\varepsilon_w = \begin{cases} \dfrac{\phi_P}{(K+K_0)^\eta}\left[1-\sqrt[3]{1-(V_{cs}-V_{cs0})}\right] & \text{for RH} = 1 \\ \dfrac{\phi_P}{(K+K_0)^\eta} & \end{cases}$$

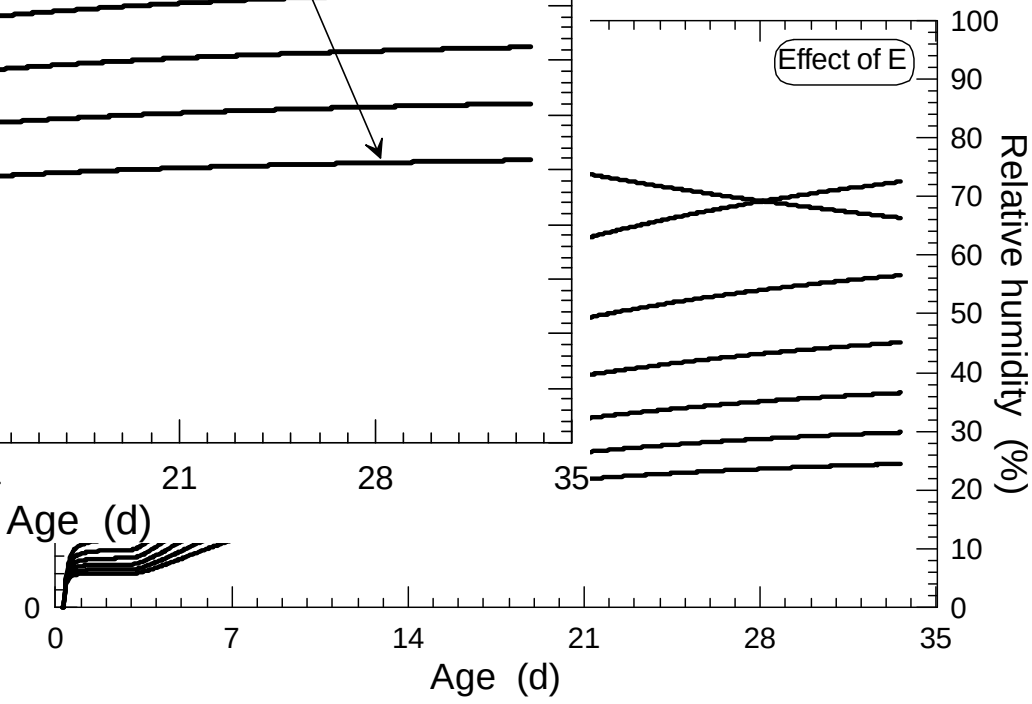
$$\xi = E / E_r \approx K / K_r$$

$$= \left(\frac{1+K_0/K_r}{\xi+K_0/K_r}\right)^\eta \quad for\ RH = 1$$

$$= \frac{1-K/K_s}{\xi-K/K_s} \quad for\ RH < 1$$

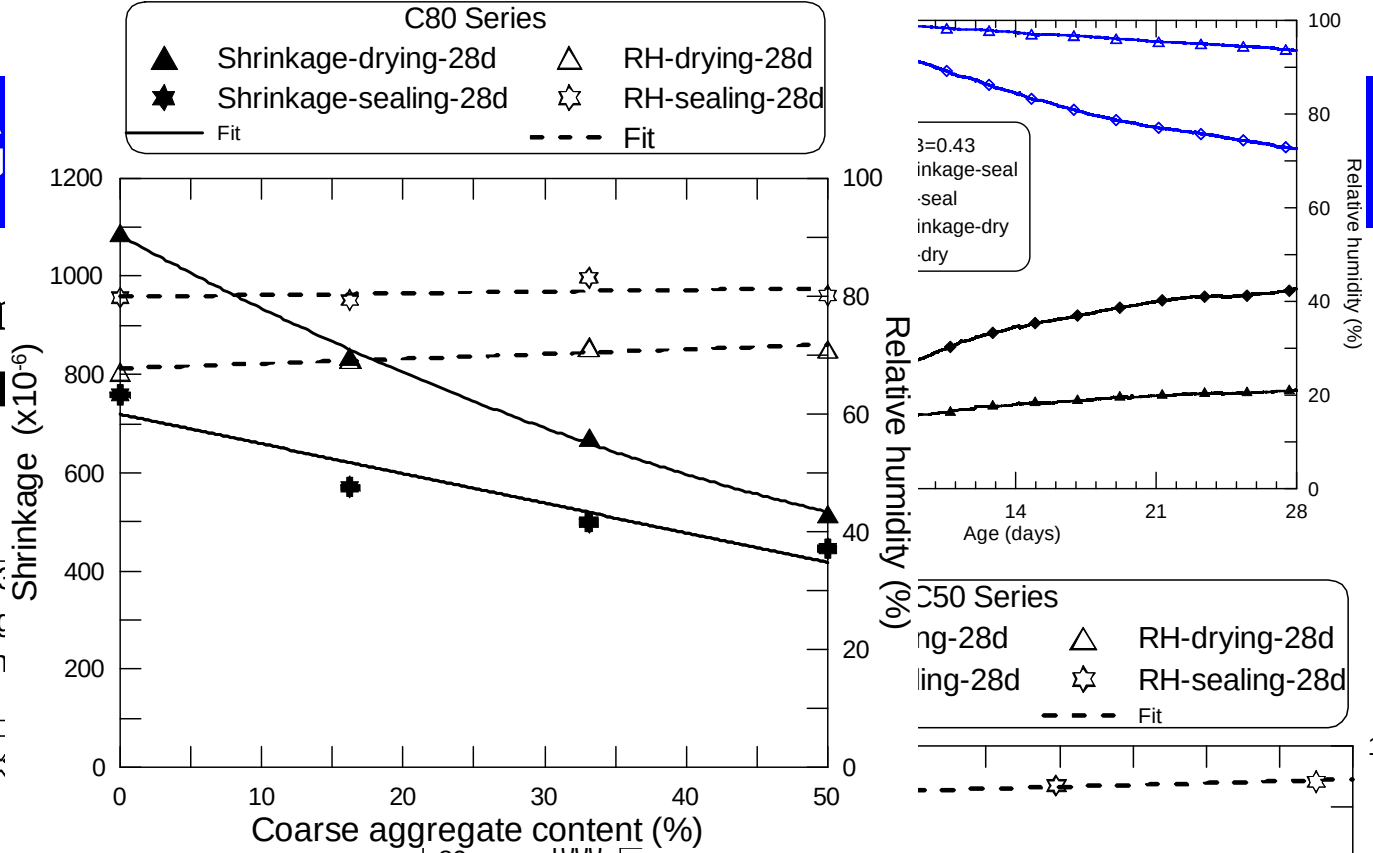


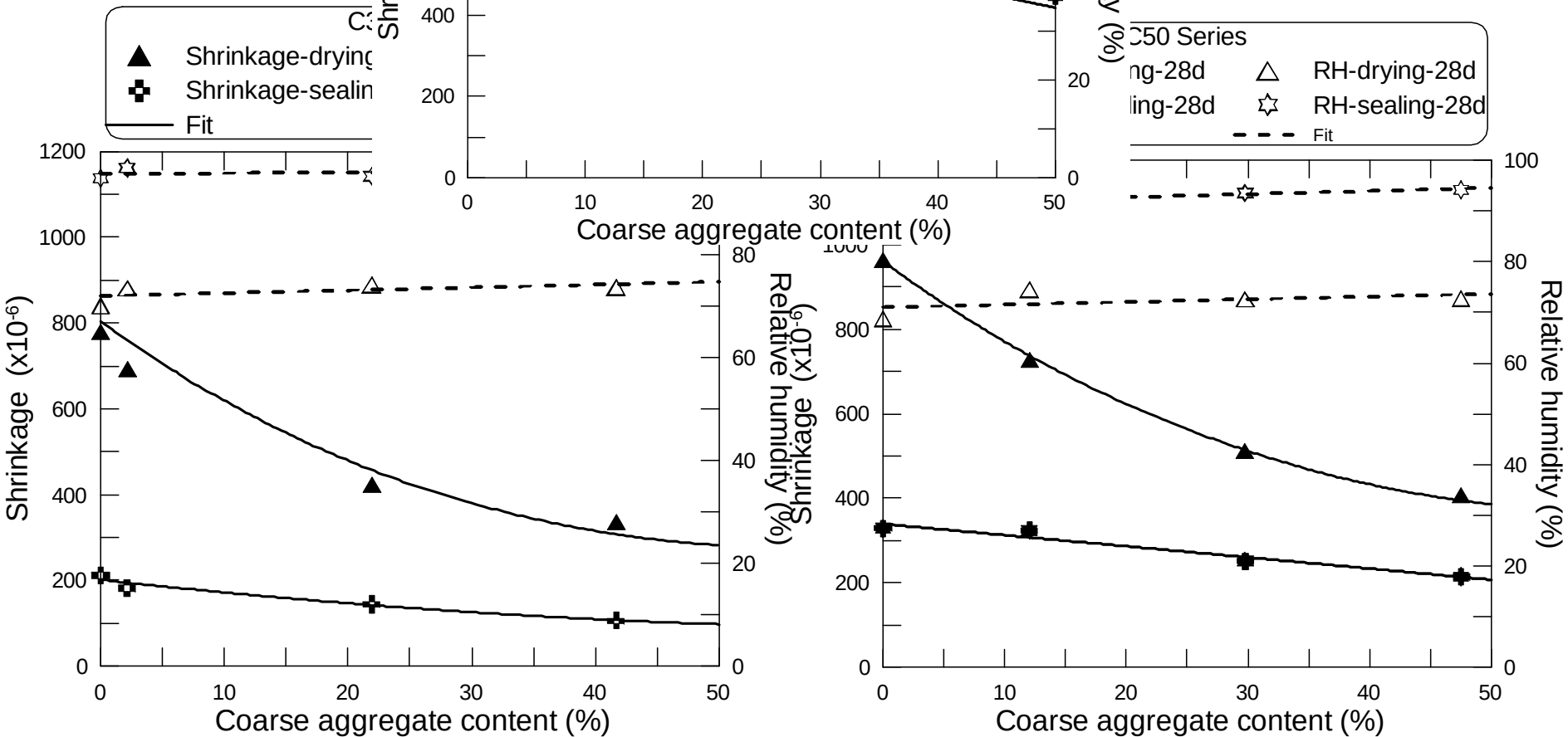




4. REG

- **Some exan**
  reduction-l

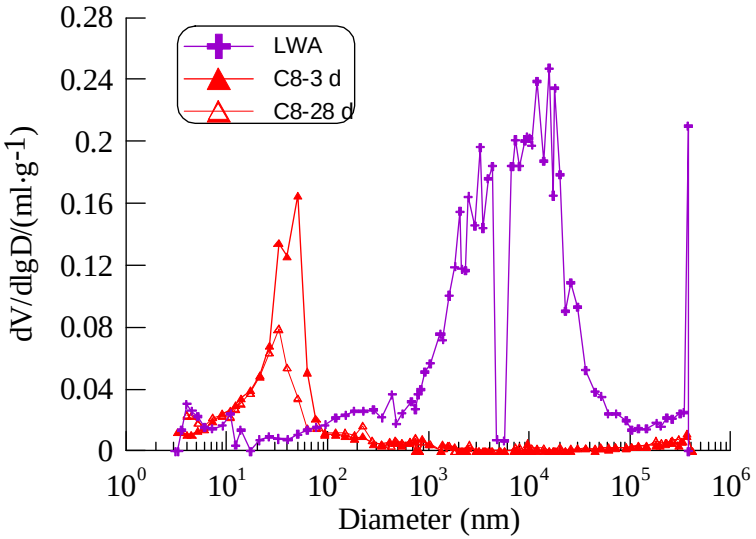




4. REGULATION

❖ Examples for shrinkage reduction-Effect of internal curing





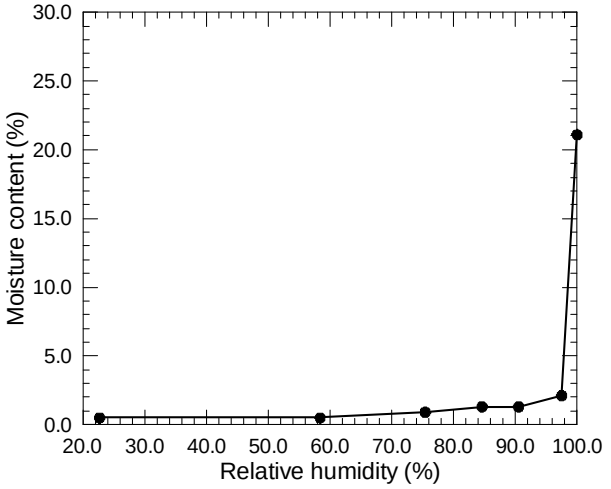
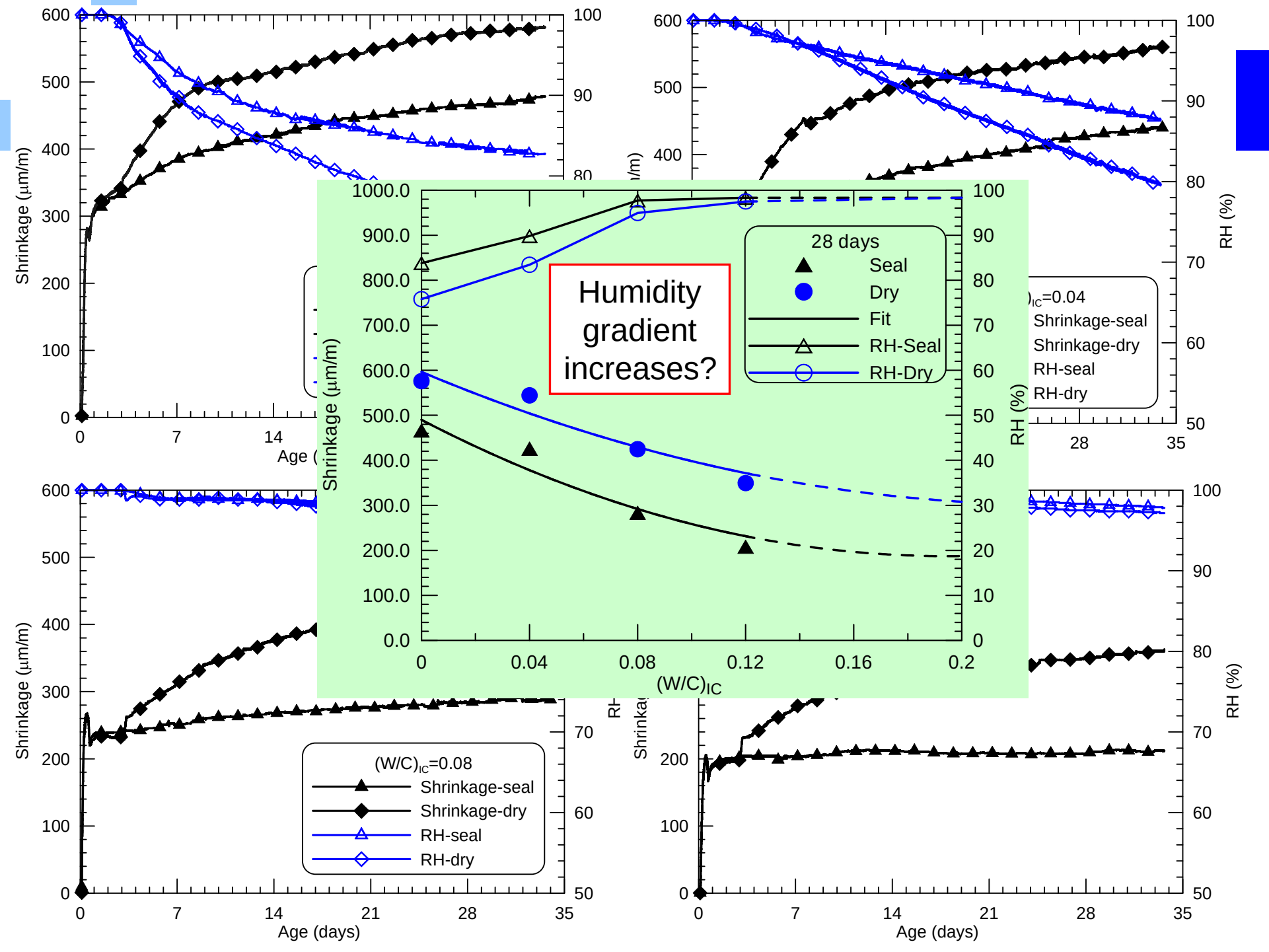


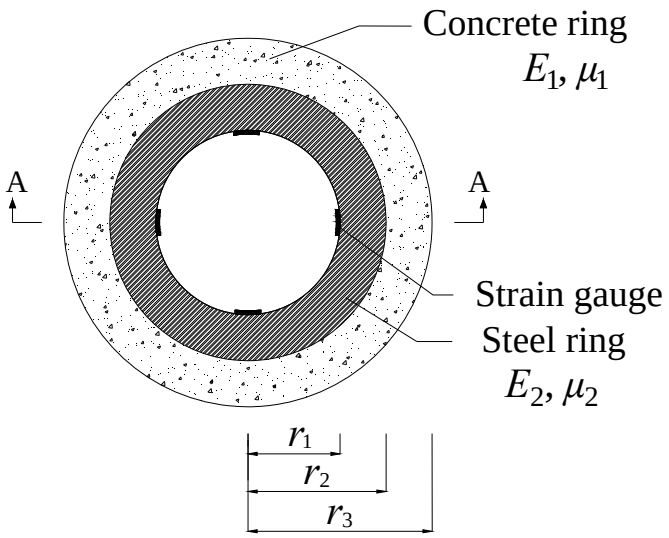
Table x. Mix proportions of concrete, kg/m3

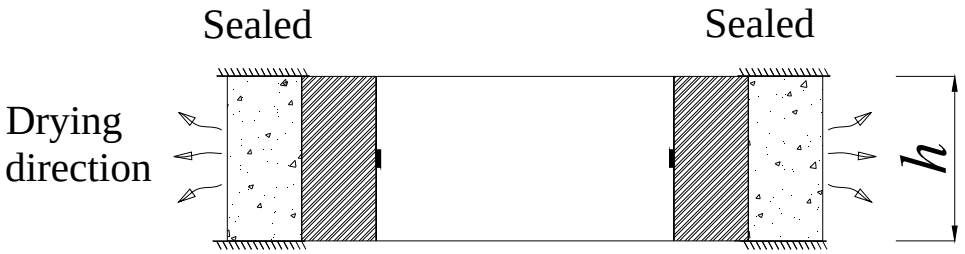
No.	Cement	Water	Sand	Stone	Silica Fume	Dry LWA	Water provided by LWA	Super-plasticizer	W/C	W_{IC}/C
C80-0	450.0	150.0	580.0	1140.0	50.0	0	0	7.47	0.333	0.00
C80-1	455.4	148.0	496.7	1055.2	50.6	86.4	18.2	7.87	0.325	0.04
C80-2	468.8	143.0	408.6	965.3	52.1	177.9	37.5	8.33	0.305	0.08
C80-3	494.4	133.5	308.9	863.7	54.9	281.3	59.3	8.85	0.270	0.12

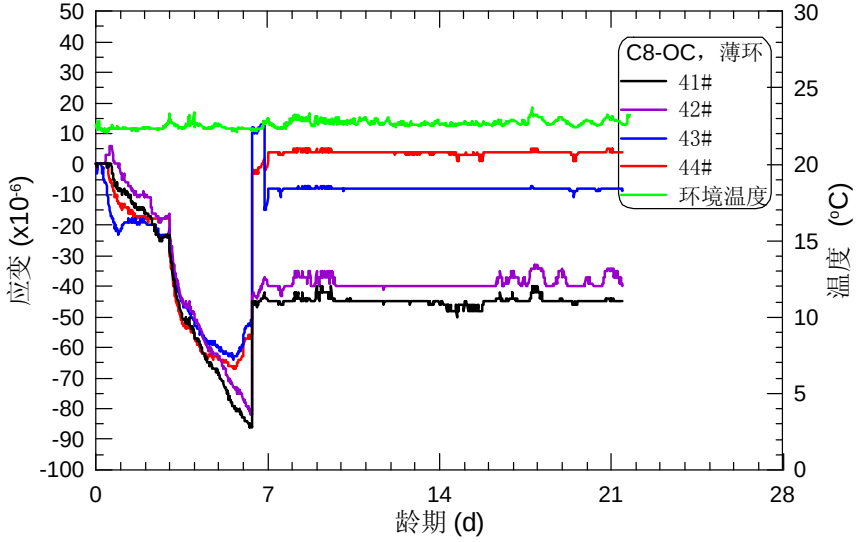

Shrinkage (µm/m)
RH (%)
Age (days)
(W/C)IC=0.08
Shrinkage-seal
Shrinkage-dry
RH-seal
RH-dry
(W/C)IC=0.04
Shrinkage-seal
Shrinkage-dry
RH-seal
RH-dry
28 days
Seal
Dry
Fit
RH-Seal
RH-Dry
Humidity gradient increases?
(W/C)IC

4. REGULATION

❖ Evaluation of Shrinkage Induced Cracking

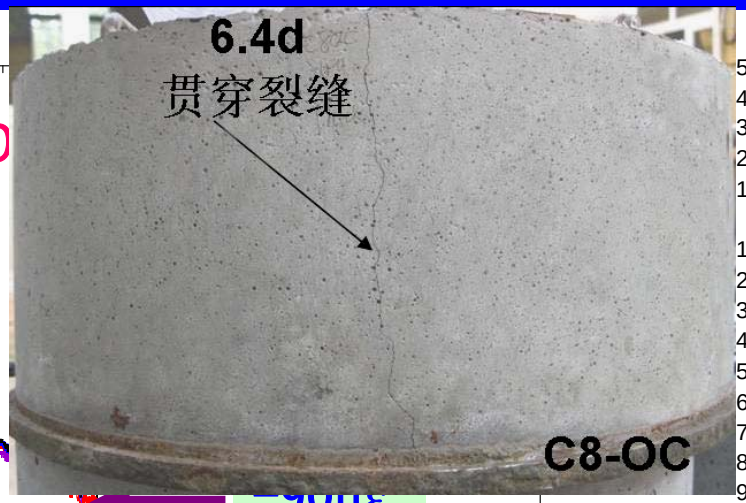


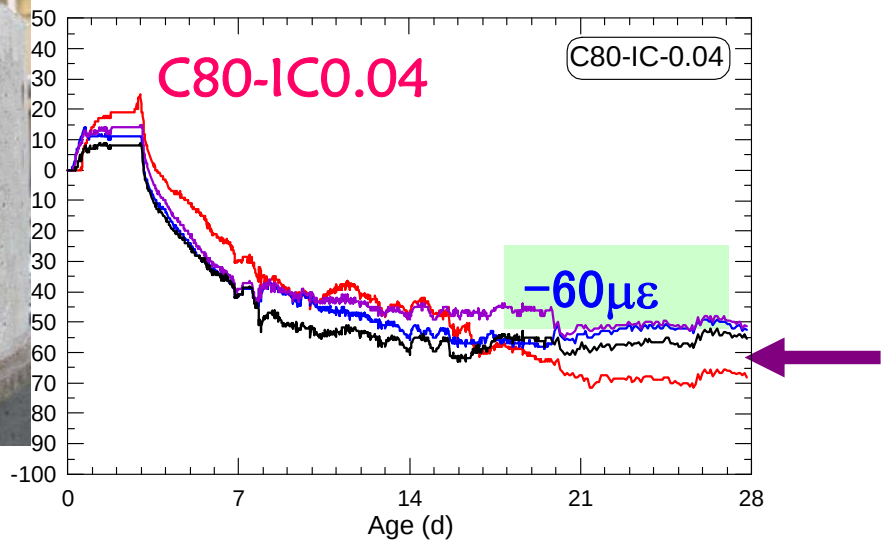


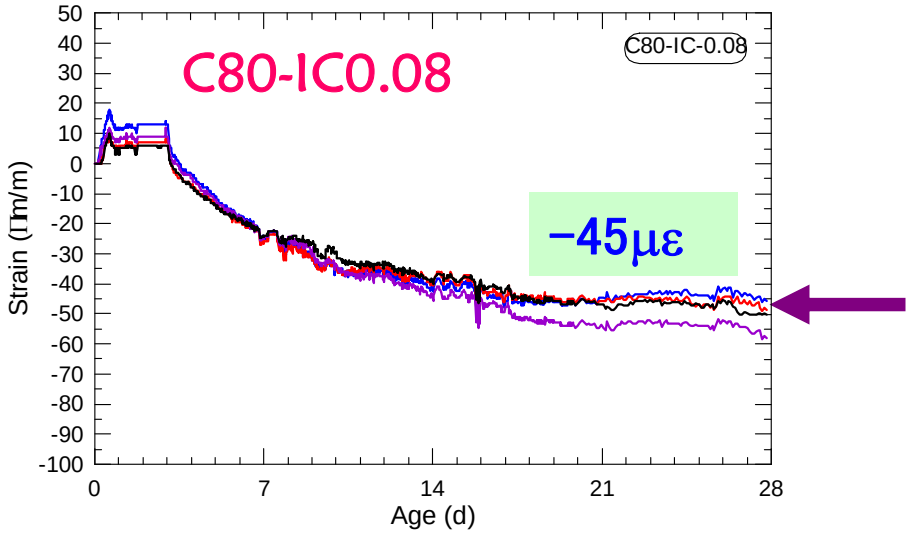


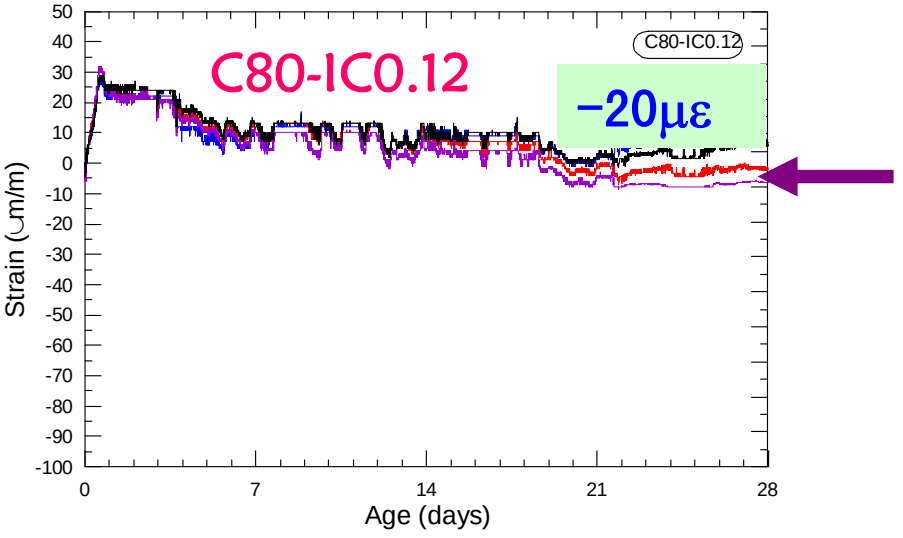


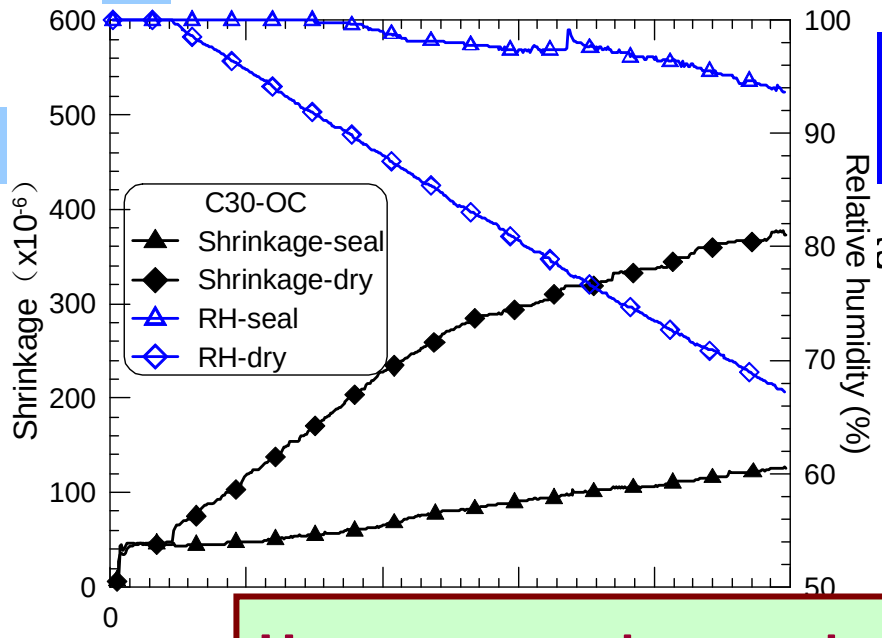
4. REGULATION



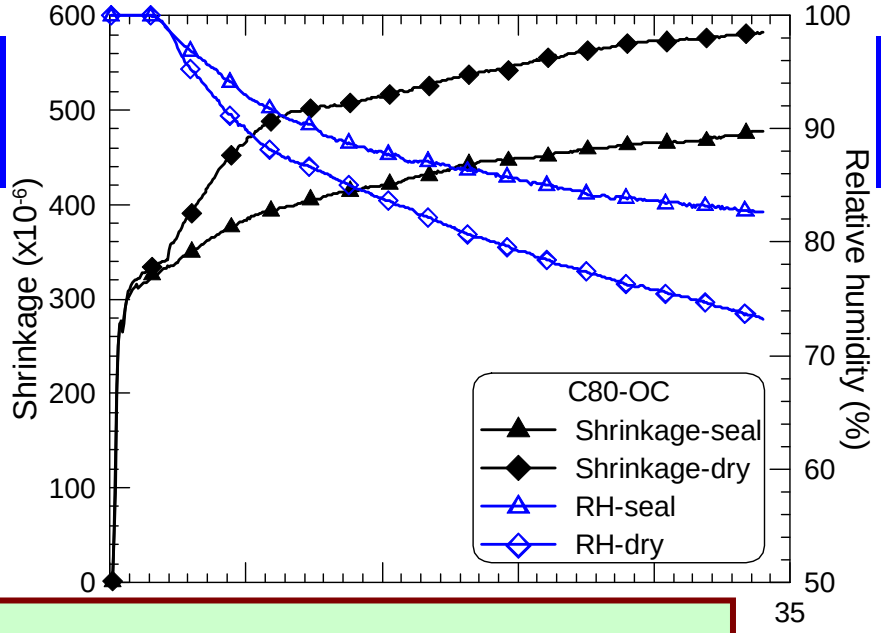




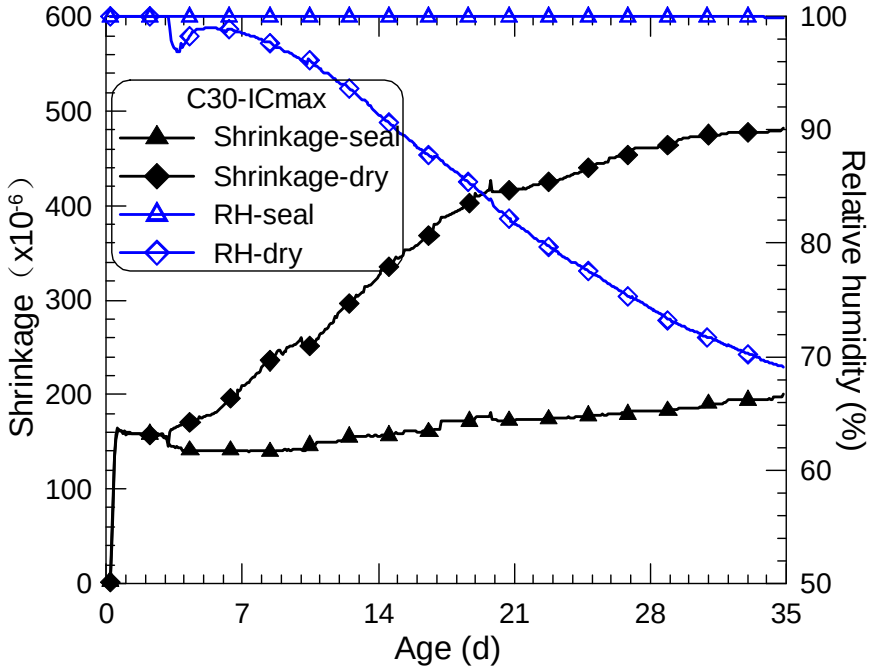


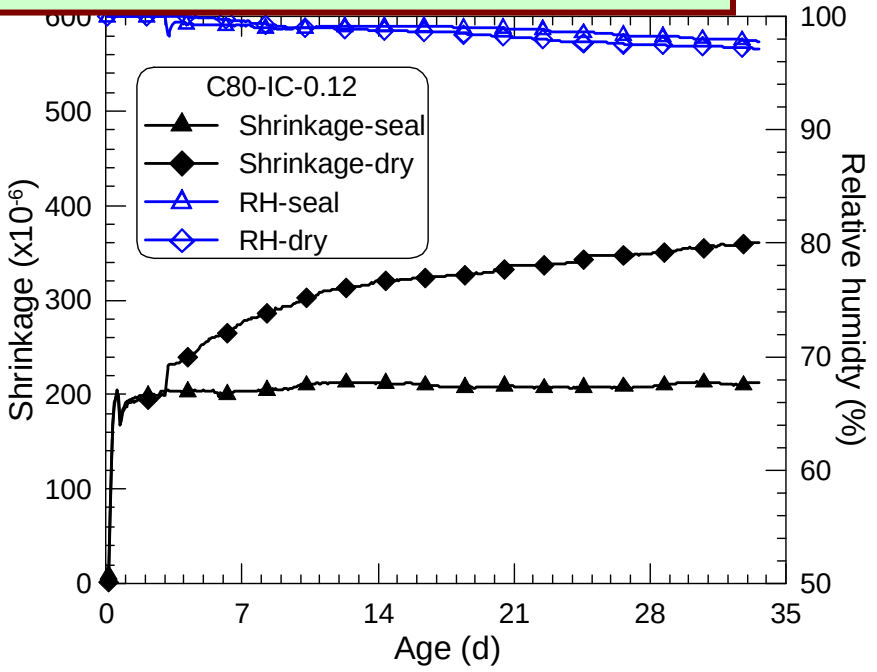


age



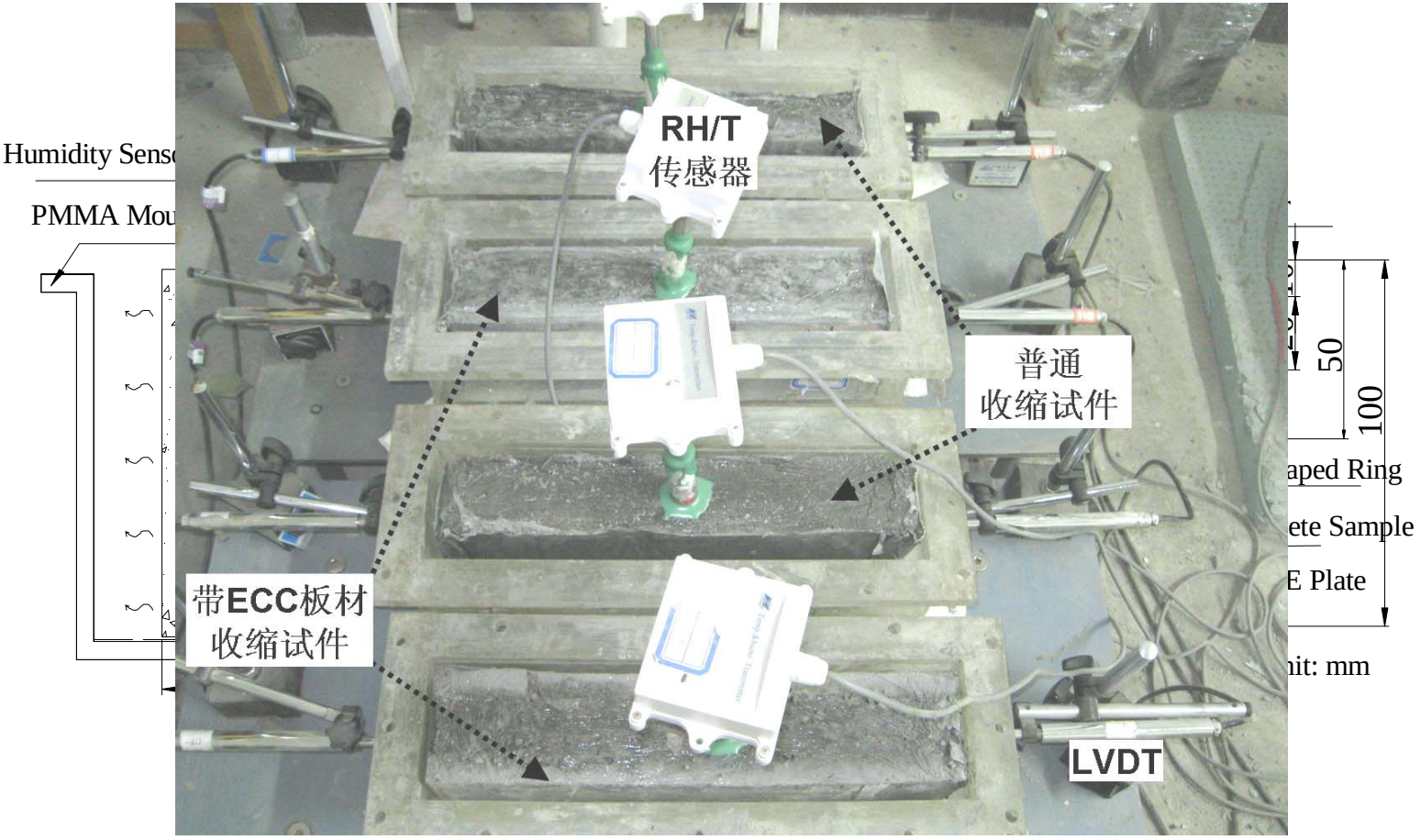
How to reduce the drying shrinkage?

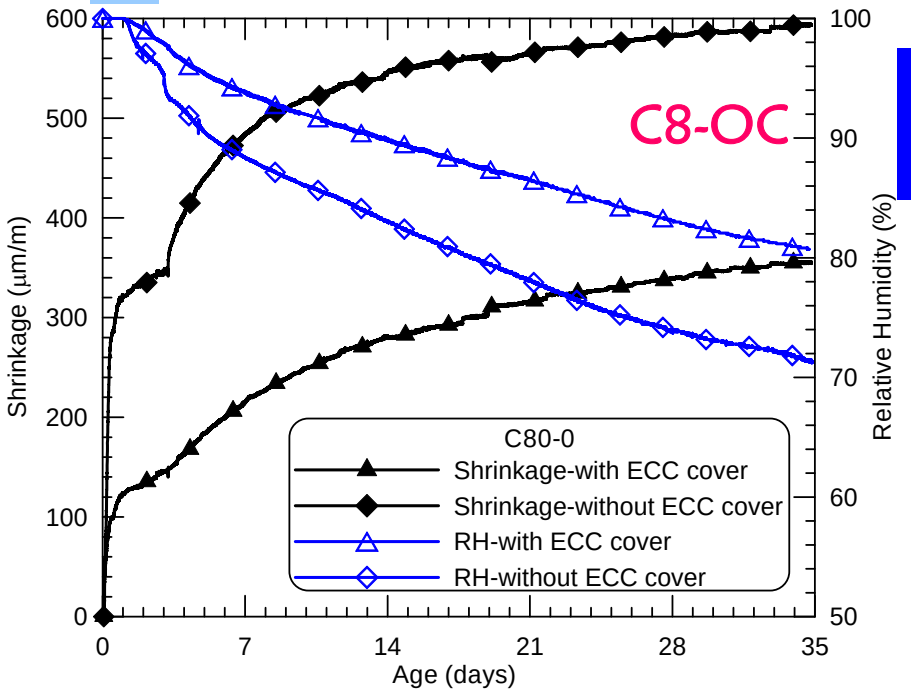


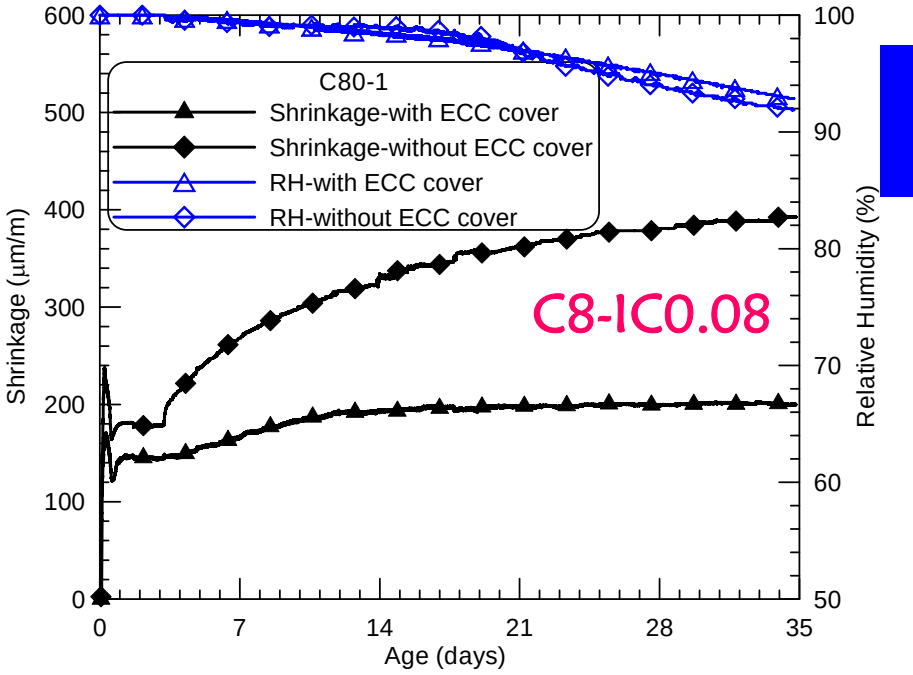


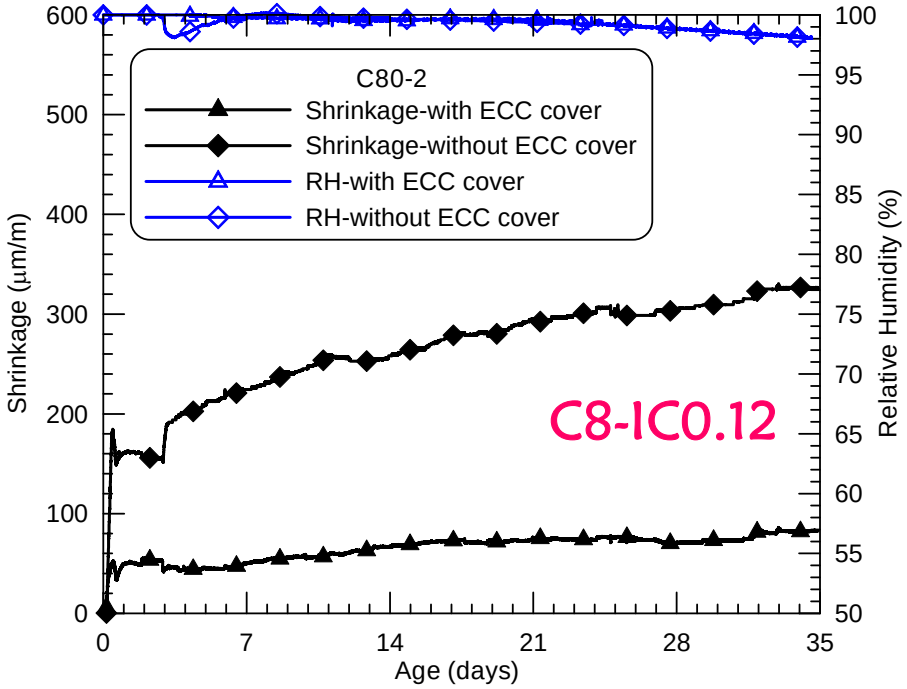
4. REGULATION

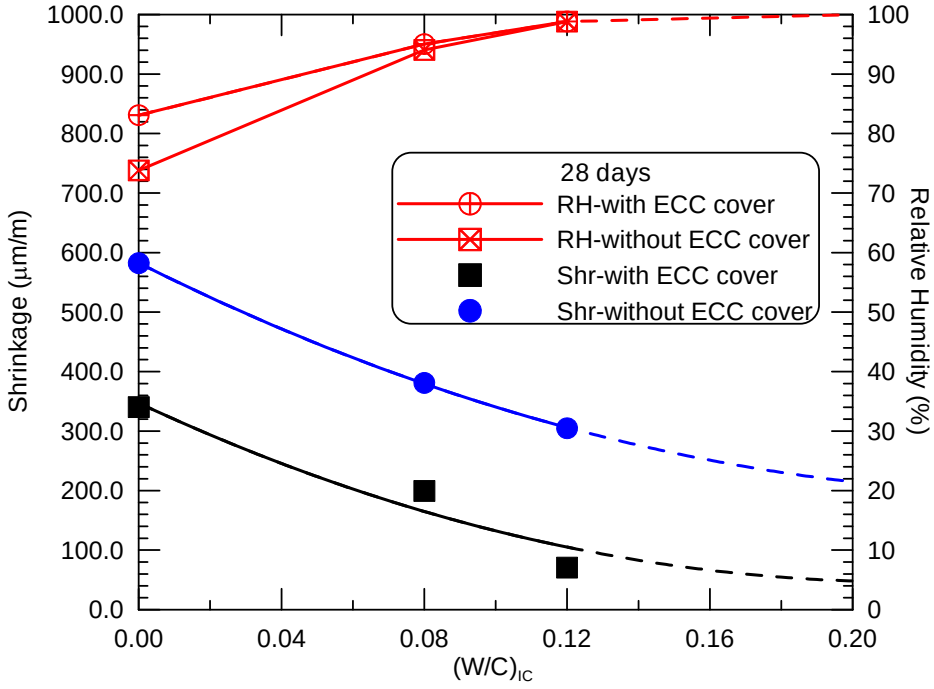
- Influences of Internal Curing and Permanent Formwork




C8-OC
C80-0
Shrinkage-with ECC cover
Shrinkage-without ECC cover
RH-with ECC cover
RH-without ECC cover
Shrinkage (μm/m)
Relative Humidity (%)
Age (days)


C80-1
Shrinkage-with ECC cover
Shrinkage-without ECC cover
RH-with ECC cover
RH-without ECC cover
C8-IC0.08
Shrinkage (μm/m)
Relative Humidity (%)
Age (days)

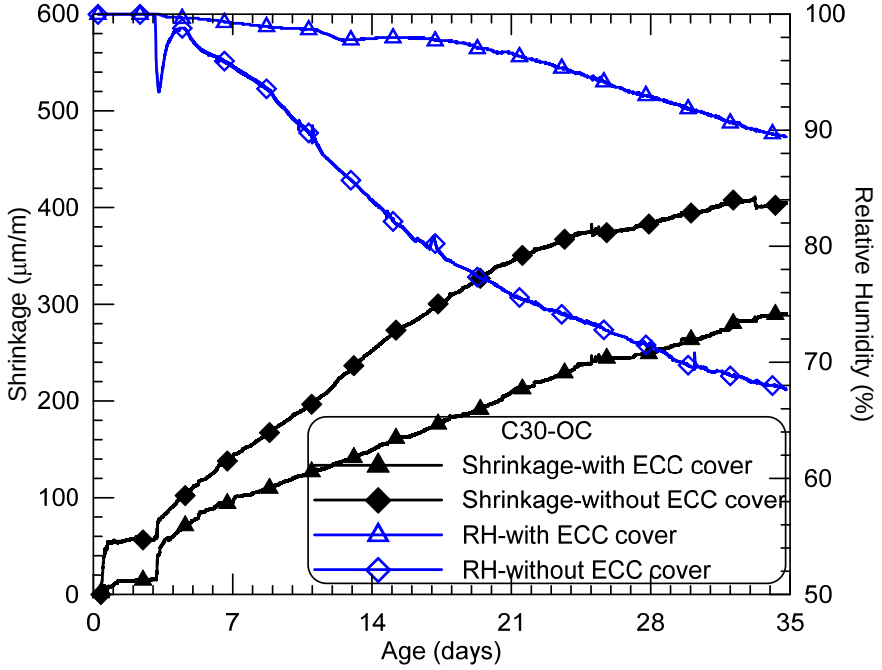

C80-2
Shrinkage-with ECC cover
Shrinkage-without ECC cover
RH-with ECC cover
RH-without ECC cover
C8-IC0.12
Shrinkage (μm/m)
Relative Humidity (%)
Age (days)


28 days
RH-with ECC cover
RH-without ECC cover
Shr-with ECC cover
Shr-without ECC cover
Shrinkage (μm/m)
Relative Humidity (%)
(W/C)IC

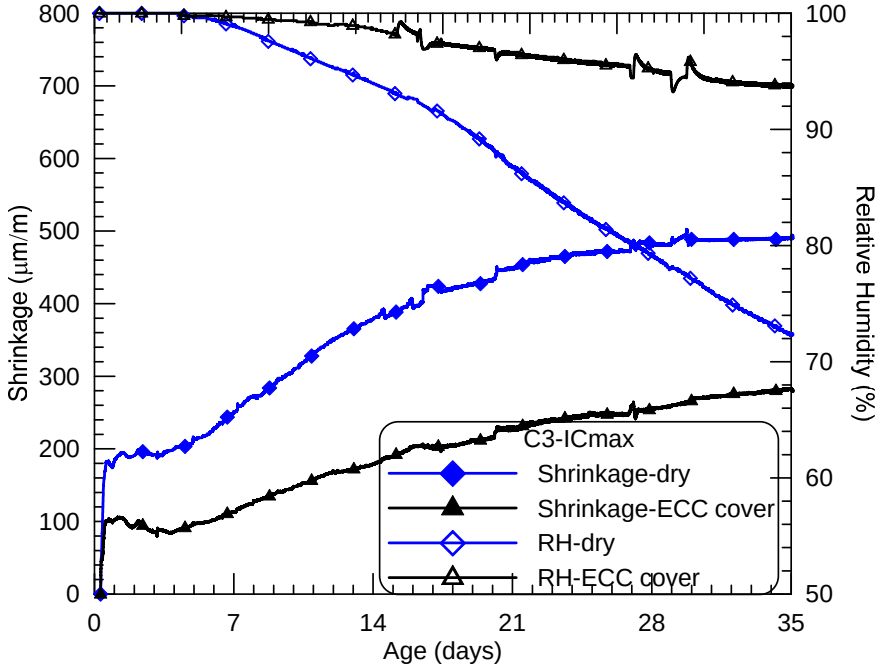
4. REGULATION

❖ Influences of Internal Curing and Permanent Formwork

C30-Concrete



OC



ICmax

5. SUMMARY AND CONCLUSIONS

- The development of internal relative humidity (RH) of concrete obeys a two-stage mode, vapor saturated stage followed by a gradually reduced stage of relative humidity.
- The development of shrinkage of concrete since set obeys a two-stage mode as well, first initial fast growth stage followed by a gradually increased stage.
- The developed model is based on chemical shrinkage of cement hydration in humidity developing stage I and capillary tension created in capillary pores in concrete in humidity developing stage II respectively.
- The model can be used for shrinkage prediction of concrete, whatever the shrinkage is caused by self-desiccation or by dying. In addition, the model can predict shrinkage strain in concrete structures not only for different time, but also for different positions.
- Model analyses show the effects of surface tension of water, length of moisture saturated stage and elastic modulus of concrete on shrinkage of concrete are significant.

ACKNOWLEDGEMENT

- This work has been supported by a Specialized Research Fund for the Doctoral Program of Higher Education (20100002110016);
- A grant from the National Science Foundation of China (No. 50978143; 51178248);
- A grant from National Basic Research Program of China (No. 2009CB623200) to Tsinghua University.
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Thank You !